SRSF-GAN: A Super-Resolution-Based Spatial Fusion With GAN for Satellite Images With Different Spatial and Temporal Resolutions

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***Abstract*— Recently, spatiotemporal fusion technologies have been rapidly developed and widely applied, which generally require one or more pairs of coarse- and fine-resolution images as reference data and a coarse-resolution image at the prediction time to produce a fine-resolution image at the forecast time. Consequently, most spatiotemporal fusion methods are phase-based and obtain temporal changing information from the coarse-resolution image pairs. Consequently, they usually have rigid constraints on reference data selection using the temporal interval criterion. However, due to the relatively long revisit cycle and cloud contamination, it is difficult to prepare adequate high-quality reference data with little change, especially for large-scale fusion tasks. Therefore, we propose a spatial-based fusion method that only requires the coarse images at the prediction time and a fine reference image selected by a spatial–spectral-similarity criterion, named super-resolution-based spatial fusion with the generative adversarial network (SRSF-GAN). SRSF-GAN uses a super-resolution (SR) module merely on the coarse images at the prediction time and then performs a multiscale fusion with the reference fine image. Moreover, the spatial attention mechanism is adopted to achieve dynamic weight tuning, i.e., assigning more weight to the SR image for changed areas and more weight to the fine reference image for unchanged areas. Comparison experiments based on three datasets show that our model can outperform the state-of-the-art methods, and the changes with different spatial and spectral ranges of variation can be recovered. The code will be uploaded to the following website: https://github.com/Zhaosir996/SRSFGAN.**

***Index Terms*— Generative adversarial network (GAN), multilevel feature, remote sensing, spatiotemporal fusion, super-resolution (SR).**

I. Introduction

REMOTE sensing time series has been widely used for studying global land surface dynamics in various fields in recent years, such as monitoring forest disturbances [1], managing agriculture changes [2], detecting urbanization [3], and providing valuable information for resource inventory and policy-making [4]. Many land cover and land use changes (i.e., the changes caused by wildfire, ice frozen/melting, flood, and rapid growth cycle of plants) can occur at multiple temporal and spatial scales, so there is a strong demand for satellite images with both high spatial and temporal resolution for high-accuracy monitoring. However, tradeoffs in sensor designs that balance spatial detail with concerns for swath width and repeat coverage always exist [5]. For example, Landsat images have a fine spatial resolution of 30 m but are with a long revisit cycle of ∼16 days. On the contrary, Moderate Resolution Imaging Spectroradiometer (MODIS) images have a frequent coverage of ∼1 days but have a coarse spatial resolution of 250 m∼1 km. Though the Sentinel data have a fine spatial resolution (10∼20 m) with a shorter revisit cycle (∼5 days with the twin satellites together), due to cloud and shadow contamination, it generally requires more than five days (probably one month) to acquire a clear image. Therefore, it is highly desirable to develop techniques to derive remote sensing data with both high spatial and temporal resolutions from current available remote sensing data.

With years of development, a large number of spatiotemporal fusion algorithms have been proposed to generate synthesized data with both high spatial resolution and frequent coverage. Moreover, these synthesized data have been successfully applied for various applications, such as crop growth monitoring [6] and disturbance detection [7]. Generally, the current spatiotemporal data fusion methods can be divided into four categories: weight-based, unmixing-based, mixture-based, and learning-based.

The weight-based models assume that the fine image can be predicted by the weighted-sum strategy according to spatiotemporal changes of coarse images. The most representative one is the spatial and temporal adaptive reflectance fusion model (STARFM) [8], which weight pixels by a

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temporal change, spectral difference, and distance within a sliding window. Based on STARFM, many improved versions of STARFM have appeared, e.g., an enhanced STARFM (ESTARFM) [9], a modified version of ESTARFM (mESTARFM) [10], and Fit-FC [11]. The weight-based models are stable and fast, yet they are easy to introduce noise and their prediction ability in heterogeneous regions is poor.

The unmixing-based models are based on the linear spectral mixing theory to reconstruct the fine-resolution image. In this category, the multisensor multiresolution technique (MMT) [12] is the first one. Others include the spatial–temporal data fusion approach (STDFA) [13] and the blocks removed spatial unmixing (SU-BR) [14]. This kind of method is physical-based and can reconstruct details to some extent. However, they usually assume that the abundance is temporally invariant, which is reasonable for many applications.

By combining the weight- and unmixing-based models, several mixture-based fusion methods are proposed, including the flexible spatiotemporal data fusion (FSDAF) [15], an enhanced FSDAF that incorporates subpixel class fraction change information (SFSDAF) [16] and FSADF2.0 [17].

Learning-based methods, which need observation data to train the model to find the fusion relationship, can be further classified into Bayesian methods [18], [19], [20], sparse representation methods [21], [22], [23], and deep learning (DL) methods [24], [25], [26], [27], [28], [29]. In recent years, due to the nonlinear representation ability, more and more DL fusion models have been proposed through the design of network structure and loss function. Initially, the deep convolutional neural network (CNN) has been introduced to the fusion problem, and both the accuracy and the adaptability of the model are significantly promoted. Thus, a series of CNN-based fusion methods have been developed since then, such as spatiotemporal satellite image fusion using deep CNNs (STFDCNN) [24], a two-stream CNN for spatiotemporal image fusion (StfNet) [25], and enhanced deep convolutional model for StfNet (EDCSTFN) [26]. Recently, since generative adversarial network (GAN)-based methods have made great success in plenty of fields, such as image generation [30], image denoising [31], and super-resolution (SR) [32], it has also been applied to spatiotemporal fusion, such as remote sensing image spatiotemporal fusion using a GAN (STFGAN) [33], a flexible reference-insensitive spatiotemporal fusion model for remote sensing images using conditional GAN (GANSTFM) [27], spatiotemporal fusion via CycleGAN-based image (CycleGAN-STF) [28], and a multilevel feature fusion with GAN for spatiotemporal remote sensing images (MLFF-GAN) [29]. GAN-based methods have shown very promising performance in spatiotemporal fusion. Recently, transformer-based spatiotemporal fusion models have been proposed and can also achieve good results, such as remote sensing spatiotemporal fusion using swin transformer (SWINSTFM) [34] and enhanced multistream remote sensing spatiotemporal fusion network based on transformer and dilated convolution (EMSNet) [35].

As we can see, all the above methods need one or more observed pairs of coarse- and fine-resolution images as reference data and another coarse-resolution image at the prediction time as input. According to the number of input images, they can be classified into two groups.

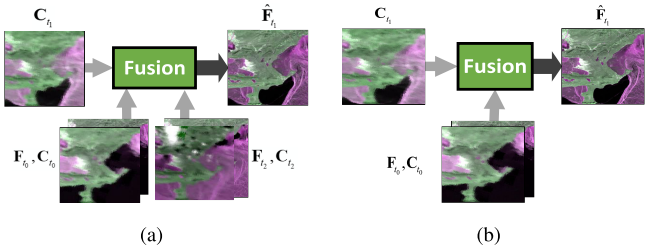


Fig. 1. Two spatiotemporal fusion patterns. (a) Two-phase reference image pattern. (b) Single-phase reference image pattern.

1) *Two-Phase Reference Image Pattern:* As shown in Fig. 1(a), this pattern requires two pairs of coarse- and fine-resolution images at two separate times, noted t_0 and t_2, and one coarse-resolution image at the prediction time, noted t_1, which can be expressed as

$$\widehat{\mathbf{F}}_{t_1} = f(\mathbf{C}_{t_1}; \mathbf{C}_{t_0}, \mathbf{F}_{t_0}, \mathbf{C}_{t_2}, \mathbf{F}_{t_2}) \tag{1}$$

where $\mathbf{C}$ and $\mathbf{F}$ represent the coarse- and fine-resolution images, respectively. $f(\cdot)$ is the spatiotemporal fusion method, and $\widehat{\mathbf{F}}$ is the predicted fine-resolution image. Generally, $t_0 < t_1$ and $t_2 > t_1$, indicating that one pair before and one pair after the prediction time are required as input. ESTARFM [9], STFDCNN [24], STFGAN [33], StfNet [25], CycleGAN-STF [28], and EMSNet [35] belong to this kind of pattern. It is worth noting that CycleGAN-STF [28] does not require coarse-resolution images at t_0 or t_1 in the DL training part.

2) *Single-Phase Reference Image Pattern:* As shown in Fig. 1(b), only one pair of coarse- and fine-resolution images is required, which is

$$\widehat{\mathbf{F}}_{t_1} = f(\mathbf{C}_{t_1}; \mathbf{C}_{t_0}, \mathbf{F}_{t_0}). \tag{2}$$

The typical algorithms are STARFM [8], Fit-FC [11], FSDAF [15], EDCSTFN [26], GANSTFM [27], MLFF-GAN [29], and SWINSTFM [34].

It can be seen that both phase-based patterns use spatial information from the input fine-resolution image and temporal information from the coarse-resolution images. That is, they assume that the spatial structure of land covers can be captured by fine-resolution reference images, while the temporal changes can be extracted from coarse-resolution images. This assumption holds for homogeneous areas or large-sized fields with gradual phenological and/or land cover changes. However, the coarse-resolution image pair is not able to accurately reflect the change characteristics of each land cover type within a coarse pixel for highly heterogeneous landscapes due to the spectral mixing problem. In addition, for the abrupt land cover type changes, phased-based methods also calculate the spatial structure change information depending on coarse-resolution images. As a result, they are not able to predict details (such as edges and broken internal texture) of abrupt changes with high precision.

Moreover, the phase-based patterns have rigid constraints on input image selection. First, the temporal interval between

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t_1 and t_0 and/or t_2 should be as small as possible, and there should be no significant ground changes during this period; second, both the reference images and $\mathbf{C}_{t_1}$ should be with high quality. The above two constraints are not easy to satisfy for practical applications, especially for the fusion task with a continuous time period and a large scale. As for the fine reference image, take the most frequently used Landsat data as an example. Due to the cloud, shadow contamination, and other factors (e.g., SLC-off problem in Landsat 7 data), the temporal interval of a clear fine reference image to t_1 is usually larger than the revisit time of the fine-resolution image and even can be several months for areas with high-frequency cloud cover. Consequently, there is a large possibility that the land cover at the reference time is different from that at the prediction time in both spatial and spectral dimensions.

For the large-scale data fusion task, we should, therefore, alleviate the strong time dependence for the reference image selection and reduce the number of input images. Essentially, using the temporal interval criterion is to find fine reference images that are as similar as possible to the ground truth in both spatial and spectral dimensions. As discussed above, due to the revisit cycle and cloud contamination, it is of great possibility that we will not able to find high-quality fine reference images without changes when using the temporal interval criterion. As a result, in this article, we abandon the phase-based fusion patterns (i.e., two-phase-based and single-phase-based) and propose a spatial-based pattern. Unlike the phased-based one, which needs the coarse image pairs to capture the temporal change information, the spatial-based one only requires the coarse images at t_1 and a fine reference image, which, therefore, has greatly reduced the requirement on data selection. Tan et al. [27] have proposed a similar way by selecting an arbitrary fine reference image. Yet, its performance will be severely affected when the fine reference image contains large changes. Therefore, in this article, we propose a new criterion for fine reference image selection, which is mostly based on the spatial and spectral similarities between the fine reference image and the coarse image at t_1. Different from the restriction on the temporal interval of phase-based pattern, we expand the search range of acquisition time so as to involve as many candidate fine images as possible and then search for the fine image with the largest similarity as the fine reference image. It can be expected that the fine image selected by our new criterion will contain fewer changes than that selected by the temporal interval criterion. Accordingly, an end-to-end network based on GAN by combining an SR and fusion module is proposed, named SR-based spatial fusion with GAN (SRSF-GAN). The SR module can directly predict the fine-resolution image at t_1, and the change information can be obtained by combining the predicted SR and the fine reference image. Therefore, coarse reference images are not required in the spatial-based pattern. Moreover, using the predicted SR image, our model can capture more fine-scale changes than the phase-based models. Next, a fusion module is designed to perform a multiscale fusion of the fine reference and SR images with weights automatically generated by a spatial attention mechanism. Since the SR process is independent of the fine-resolution image, our model is able to correctly predict multiscale changes by putting more weight on the SR image for changed areas through the spatial attention mechanism.

The rest of this article is organized as follows. The detailed architecture of the proposed model is introduced in Section II. In Section III, comparison and ablation experiments are presented in detail. The conclusion and future work are summarized in Section IV.

II. Methodology

For real applications, it is usually difficult to find all coarse- and fine-resolution images with high quality, especially when the temporal and spatial scale is large. Therefore, we abandon the phased-based patterns that use the temporal interval criterion to search the fine reference image. In this article, we apply the spatial-based pattern and adopt a new criterion to find the fine reference image, named the spatial–spectral-matched (SSM) criterion. That is, we search from a set of archived fine-resolution images and select the one with the largest spectral–spectral similarity to the coarse images at t_1. As a result, this pattern has largely loosed the temporal restriction on the reference image selection and, thus, greatly reduced the requirement for the input data. Moreover, we can find more similar fine reference images than phased-based patterns. The overall flowchart of our method is shown in Fig. 2, which mainly includes two parts.

1) *Data Selection:* In this article, we use the 30-m Landsat data to represent the fine-spatial-low-temporal image and the 250-m MODIS data to represent the high-temporal-coarse-spatial image. Theoretically, if the quality of the coarse image is fine and stable, there is no need for data selection. However, we use the MODIS data here, which has several data quality problems, so we also design a selection criterion for it. Two coarse images around t_1 with high quality and rich texture information are selected, denoted as $\mathbf{C}_{t_1}^1, \mathbf{C}_{t_1}^2$ (simplified as $\mathbf{C}_1, \mathbf{C}_2$). For the fine reference image, among p candidate images, we select the image with the highest SSM to $\mathbf{C}_1$ and $\mathbf{C}_2$ as the spectral-spatial-matched reference image, denoted $\mathbf{F}_{\mathrm{SSM}}$.
2) *Image Fusion:* A new GAN network structure is proposed for high-quality spatial fusion of the coarse images at t_1 and the fine reference image, which can be expressed as follows:

$$\widehat{\mathbf{F}} = f(\mathbf{C}_1, \mathbf{C}_2; \mathbf{F}_{\mathrm{SSM}}). \tag{3}$$

A detailed description of our method will be presented in the following.

A. Data Selection

1) Data Selection for the Coarse Images at t_1: Except for the similar cloud contamination problem as the Landsat image, the MODIS data have two extra data problems: "no valid data (NVD)" and "spatial resolution variability." The first problem is that MODIS is not able to cover some areas on some days within a year, so the corresponding images are not available for further processing (see Fig. 3). The second

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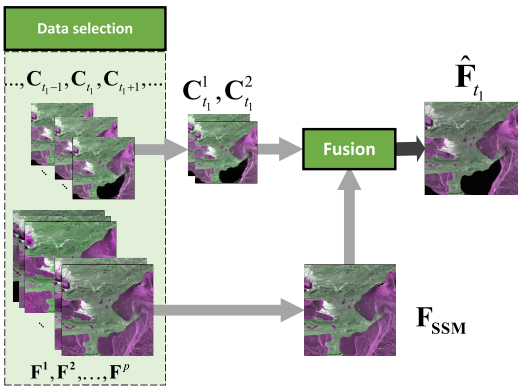
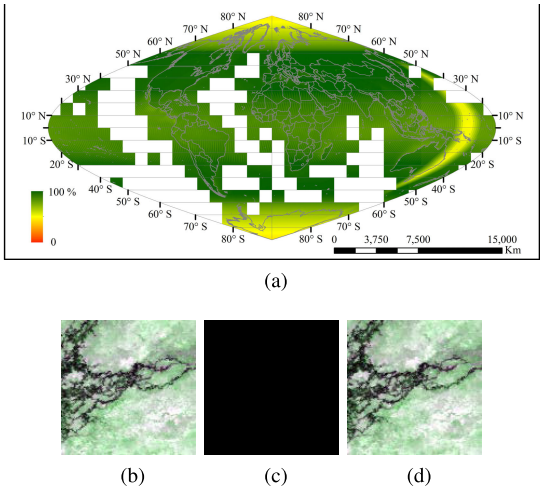


Fig. 2. Spatial-based fusion pattern of our method.



(a)

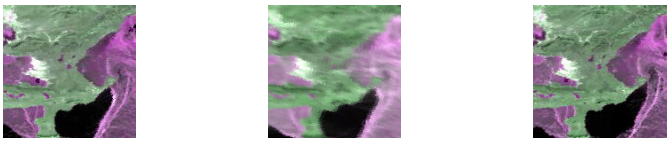
(b) (c) (d)

Fig. 3. "NVD" problem of MODIS data [36]. (a) Average percent of valid data from 2001 to 2016. (b) DOY = 163 and year = 2020. (c) DOY = 164 and year = 2020. (d) DOY = 165 and year = 2020.

problem is that the clarity of the MODIS images for different days of the same site is different (see Fig. 4) though the spatial resolution of all standard MODIS reflectance images is set to the same. It is because the footprints of MODIS observations on different days can vary considerably in size and location, yet they are all binned into the same predefined grid cell toward near real-time monitoring of forest disturbance by fusion of MODIS and Landsat data. This phenomenon can also be reflected in the recorded range layer, which is defined as the distance between the sensor and the ground (https://modis.gsfc.nasa.gov/data/dataprod/mod09.php). As shown in Fig. 5, the darker the color, the MODIS reflectance image with a higher clarity usually has a smaller range value. Due to the above two problems, data selection for the MODIS data in advance is of great necessity for the practical application. The selection criteria are designed as follows.

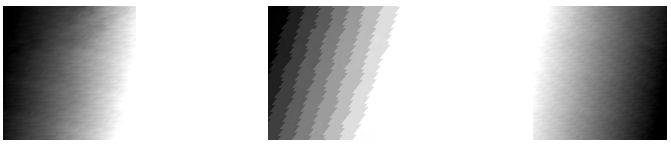
1) *Percentage of "NVD" Pixels:* $p_{\text{NVD}} \leq T_{\text{NVD}}$.
2) *Percentage of Cloud Pixels:* $p_{\text{cloud}} \leq T_{\text{cloud}}$.
3) *Temporal Interval:* $\Delta t \leq T_{\text{time}}$.
4) *Average Range of the Data:* Range $\leq T_{\text{range}}$.

Generally, the larger the thresholds T_{NVD}, T_{cloud}, T_{time}, and T_{range} are, the more candidate MODIS images can be selected. However, if the ground objects are highly dynamic,



(a) (b) (c)

Fig. 4. "Spatial resolution variability" problem of MODIS daily image data. (a) DOY = 17 and year = 2018. (b) DOY = 18 and year = 2018. (c) DOY = 19 and year = 2018.



(a) (b) (c)

Fig. 5. Range value of MODIS daily image data. (a) DOY = 17 and year = 2018; the average range is 28 190 m. (b) DOY = 18 and year = 2018; the average range is 52 235 m. (c) DOY = 19 and year = 2018; the average range is 28 190 m.

T_{time} should be set small. In this article, we set $T_{\text{NVD}} = 0.05$, $T_{\text{cloud}} = 0.05$, $T_{\text{time}} = 2$, and $T_{\text{range}} = 28\,500$. Suppose that there are p candidate MODIS images; we further select two images with the largest texture information as the input data. Here, the NIQE [37] index is used to evaluate the information of images. Two images with the largest NIQE values are selected for $\mathbf{C}_1$ and $\mathbf{C}_2$. Yet, if no coarse-resolution image can be selected by the four criteria, one can do the following.

1) Adjust the four thresholds: T_{NVD}, T_{cloud}, T_{time}, and T_{range}. However, if T_{NVD} or T_{cloud} is set large, an MODIS image with NVD or cloud pixels may be selected as input. As a result, our model will also output NVD or cloud pixels correspondingly. If T_{range} is set large, MODIS images with actual spatial resolution much larger than 250 m may be selected. Then, the quality of the predicted image may be degraded.
2) Directly use the MODIS image at t_1 as the input. Again, the output result may have the NVD, cloud, or low-quality problem if the MODIS image at t_1 has these problems.

2) Data Selection for F_{SSM}: The commonly used criterion for the reference image is the temporal interval between reference and prediction time. It is based on the assumption that two temporally close images are usually spatially and spectrally similar. However, due to the 16-day Landsat revisit cycle and frequent cloud contaminant, it is not always easy to find a fine reference image that is not only cloud clear but also temporally close to t_1. Moreover, even if the fine reference image is clear and its temporal interval is small, and if the ground objects are dramatically dynamic, lots of changes will exist on the reference image. In this case, using this fine image as the reference will largely increase the possibility of making incorrect predictions on changed areas. Therefore, the temporal interval is not an ideal criterion for selecting the reference image.

As a matter of fact, the main object of the temporal interval criterion is to find the fine reference image, which is spatially

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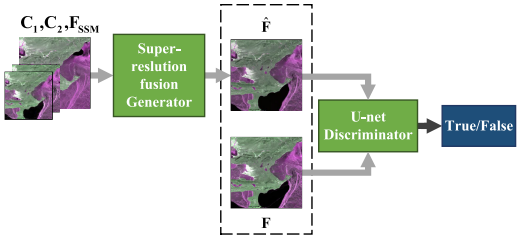


Fig. 6. Overall network of our model. The generator is responsible for generating the prediction image, and the discriminator is used to determine whether the image is true or false.

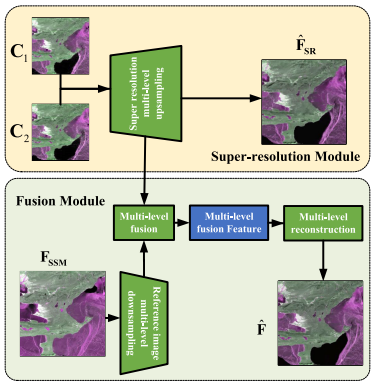


Fig. 7. Overall network of generator. The yellow part is the SR module, and the green part is the fusion one.

and spectrally similar to the coarse image at t_1. Therefore, in this article, we directly design an SSM metric to search for the optimal reference image. To find the most similar one, we first largely expand the temporal selection range. That is, we build a large candidate reference image set, which contains all the Landsat images with cloud pixels less than 5% within recent n years, denoted $\{\mathbf{F}_1, \mathbf{F}_2, \ldots, \mathbf{F}_P\}$, where P is the total number of candidate reference images. Next, we define the SSM metric for $\mathbf{F}_p$ based on the structure similarity index measure (SSIM) [38], named aSSIM, as follows:

$$\text{aSSIM}(\mathbf{C}_1, \mathbf{C}_2, \mathbf{F}_p) = \frac{1}{K}\sum_{k=1}^{K}(\text{SSIM}(\mathbf{C}_{1k}, \mathbf{F}_{pk}) + \text{SSIM}(\mathbf{C}_{2k}, \mathbf{F}_{pk})) \quad (4)$$

where the function $\text{SSIM}(\mathbf{X}, \mathbf{Y})$ is to calculate the SSIM value of $\mathbf{X}$ and $\mathbf{Y}$, $\mathbf{C}_{1k}$, $\mathbf{C}_{2k}$, and $\mathbf{F}_{pk}$ represent the kth band of $\mathbf{C}_1$, $\mathbf{C}_2$, and $\mathbf{F}_p$, respectively ($1 \le k \le K$ and K is the total number of bands). Then, the one with the highest aSSIM value is selected as the final reference image (i.e., $\mathbf{F}_{\text{SSM}}$).

B. Overall Model Design

In this article, the GAN network structure [39] is adopted. Like other GAN networks, our model also consists of two parts: the generator and discriminator, as shown in Fig. 6. The generator is used to generate the high-quality spatiotemporal fusion image, while the discriminator is to identify the generated images to guide the training of the generator.

As expressed in (3), the input coarse-fine pair may not be temporally consistent, which means that the distribution of land covers on the coarse- and fine-resolution images could be different. Although we have made efforts to guarantee that the selected reference image has the highest similarity with the coarse ones, it is still possible that some ground objects have been greatly changed spatially and/or spectrally. It can be easily found out that fusing simply the coarse- and fine-resolution images is not able to generate a satisfactory result on those changed areas. A natural solution to this problem is that, for changed areas, the prediction should depend mainly on the coarse images, while, for the unchanged areas, the prediction should rely largely on the fine reference image. Based on this idea, we propose a new spatial-based model, named SRSF-GAN (see Fig. 7). An SR module is introduced to generate a fine-resolution image merely based on the coarse images at t_1. Then, our model is able to capture the changed areas by combining the predicted SR image and fine reference image. It should be noted that the SR process is completely independent of the fine reference image, so, even when the ground objects change greatly, our model still has the ability to predict the changed areas. Next, a fusion model is further introduced to fuse the reference and predicted SR images in multiscale stages. In this step, more spatial details can be gained from the fine reference image for areas with little changes.

C. SR Module

As discussed above, the SR module is aimed to derive a fine-resolution image at t_1 merely based on the coarse-resolution images at t_1. The fine reference image is not involved here. The traditional SR methods include the nearest neighbor, bilinear, and cubic convolution upsampling, which are currently believed to recover little detailed information. Thereby, in this article, we use a deep residual network to improve the details of the upsampling image. So far, many SR methods based on DL networks have been proposed, such as image SR using deep convolutional networks (SRCNN) [40], accurate image SR using very deep convolutional networks (VDSR) [41], deeply recursive convolutional network for image SR (DRCN) [42], enhanced deep residual networks for single image SR (EDSR) [43], deep back-projection networks for SR (DPBN) [44], and densely residual Laplacian SR (DRLN) [45].

Among them, EDSR [43] has presented a satisfactory performance in most SR tasks, so, in this article, we design our SR module based on EDSR. The main idea of EDSR is removing the normalization layer from the Resblock, which is beneficial to the lightweight module and can also help to improve the quality of the generated image. The detailed network architecture of our SR model is shown in the yellow part of Fig. 8. First, we use a head convolution to initially extract $\mathbf{C}_1$'s and $\mathbf{C}_2$'s features and combine them. We then use a Resblock whose normalization is removed to extract multilevel features and deconvolution for upsampling, with a multiplicity of 2 for each level. Here, the multilevel features are denoted $\mathbf{C}_{\text{SR}}^i, i = 1 \sim M$, where M is the total number of levels. In this article, the spatial resolution difference between

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TABLE I
PARAMETERS OF BLOCKS IN THE SR MODULE

	kernel size	stride	Number of channels
Head	1	1	64
Resblock	3	1	64
Upsampling deconvolution	3	2	64
Tail	1	1	3

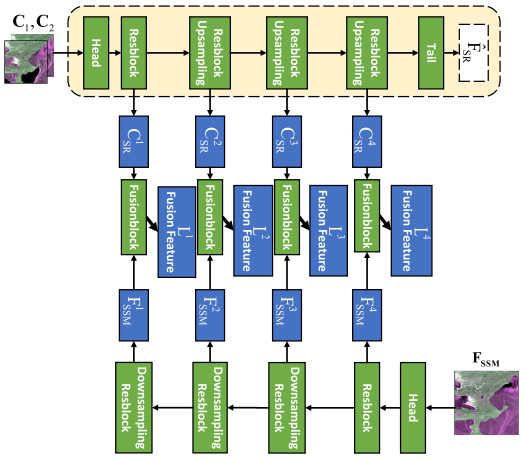


Fig. 8. Detailed structure of the SR (in yellow) and fusion modules.

the coarse and fine image is $\times 8$ times, and the upsampling scale is $\times 2$ for each level, so $M = 4$. Finally, we use a tail convolution to output the SR image so that we can use the loss function to control the SR module. We have listed the specific parameters of the convolutions in Table I; it is worth noting that the number of parameters is the same for the three resblock and upsampling blocks.

D. Fusion Module

Though the SR module can improve the image resolution, it can be expected that a lot of image details will be missed because the resolution difference of the fusion scenes (i.e., $8\times$) is up to the limitation of the SR module. Therefore, a fusion module is introduced here, which mainly includes two parts: the multilevel fusion and the multilevel reconstruction (see Fig. 2). The former is designed to generate multilevel fusion features (noted $\mathbf{L}_i$, $1 \le i \le 4$) from $\hat{\mathbf{F}}_{\mathrm{SR}}$ and $\mathbf{F}_{\mathrm{SSM}}$, while the latter is to reconstruct $\hat{\mathbf{F}}$ from $\mathbf{L}_{1\sim4}$.

1) Multilevel Fusion: Theoretically, a spatiotemporal fusion model should adopt the ground details from fine reference images at the unchanged areas while predicting ground details from the input coarse image pairs at the changed areas. Thus, there are two crucial steps for the fusing process: 1) how to predict the changing areas and 2) how to integrate the coarse and fine information in the changed areas. Generally, in the single- and two-phase patterns, the changed areas are calculated from the coarse image pair [i.e., $\mathbf{C}_{t_1}$ and $\mathbf{C}_{t_2}$ in (1) and (2)]. Yet, we are not able to derive this key information in this way since only coarse images at t_1 are utilized here. Alternatively, we escape the first step and directly fuse the coarse and fine reference images. Moreover, since the changes can happen at multiple scales, a multilevel fusion structure is proposed here, as shown in Fig. 8.

The common way for multilevel feature extraction is: 1) upscaling the coarse image to the fine resolution by the nearest neighbor or bilinear upscaling method and 2) downscaling the resized coarse and fine images using convolution to get the multiscale features from local to global. As discussed in Section II-C, few details can be recovered by the traditional upscaling method in the first step. Therefore, the multilevel features ($\mathbf{C}_{\mathrm{SR}}^{1\sim4}$) generated in the SR module are utilized for fusion. We believe that they are more accurate since we have added a term in the final loss function to improve the performance of the SR module and, thus, can reconstruct more ground truth. As for $\mathbf{F}_{\mathrm{SSM}}$, we use the same Resblock in the SR module for downscaling, and the multilevel features are denoted $\mathbf{F}_{\mathrm{SSM}}^{4\sim1}$ in the reverse order.

Then, for each level, a Fusionblock based on attentional feature fusion (AFF) [46] is proposed to fuse ith level features ($\mathbf{C}_{\mathrm{SR}}^{i}$ and $\mathbf{F}_{\mathrm{SSM}}^{i}$), as shown in Fig. 9. The original AFF uses two layers of convolution in the local and global attention blocks, and its number of weight maps is equal to the number of channels of the input features. Thus, both the computational complexity and memory requirement are high. To improve the effectiveness, a compact AFF is proposed here. First, we only keep one convolution kernel in the local and global attention blocks (named Conv1). Compared to the original AFF, the compact one has reduced one convolutional layer and one batchnorm layer. Second, all channels share the same weight map. That is, the ith level fusion feature, noted $\mathbf{L}^i$, is computed by

$$\mathbf{L}^i = M\left(\left|\mathbf{F}_{\mathrm{SSM}}^i - \mathbf{C}_{\mathrm{SR}}^i\right|\right) \otimes \mathbf{C}_{\mathrm{SR}}^i + \left(\mathbf{1} - M\left(\left|\mathbf{F}_{\mathrm{SSM}}^i - \mathbf{C}_{\mathrm{SR}}^i\right|\right)\right) \otimes \mathbf{F}_{\mathrm{SSM}}^i \quad (5)$$

where $\mathbf{F}_{\mathrm{SSM}}^i$ and $\mathbf{C}_{\mathrm{SR}}^i$ are the ith fine reference and SR features, respectively. $\otimes$ represents the pixel multiplication. $|.|$ represents the absolute value calculation. $M(.)$ defines the whole attention module, whose output corresponds to the weight map ranging from 0 to 1. It should be noted here that $M(|\mathbf{F}_{\mathrm{SSM}}^i - \mathbf{C}_{\mathrm{SR}}^i|)$ is same for all channels. The reference image tends to represent the details of the unchanged features, while the SR image can have the ability to provide changed features. According to (5), the larger the weight is, the greater the fused feature is dependent on the fine reference image (i.e., $\mathbf{F}_{\mathrm{SSM}}^i$), and vice versa. Using this compact AFF mechanism, we expect the network to be self-adjusted by enlarging the weights for areas with little changes while decreasing the weights for areas with great changes.

2) Multilevel Reconstruction: Similar to the previous SR module, a multilevel reconstruction module is proposed (see Fig. 10). First, the lowest resolution feature $\mathbf{L}^1$ is input to a DRLB and upsampling block to generate an upsampled feature. Then, it is concatenated with the second-level fusion feature $\mathbf{L}^2$ and passed through a convolution named Conv1 with a convolution kernel of 1 and a channel count of 64 to form a new feature. This process is repeated three times to complete the whole integration of different scale features. Finally, we use a tail convolution to complete reconstruction for the final fusion image $\hat{\mathbf{F}}$.

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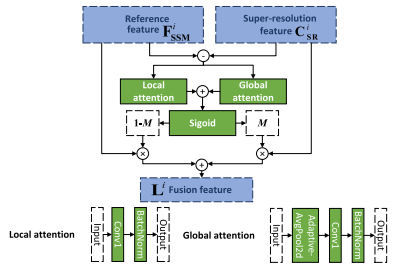


Fig. 9. Architecture of the Fusionblock that is used to fuse the fine reference and SR features at the ith level ($i = 1{\sim}4$).

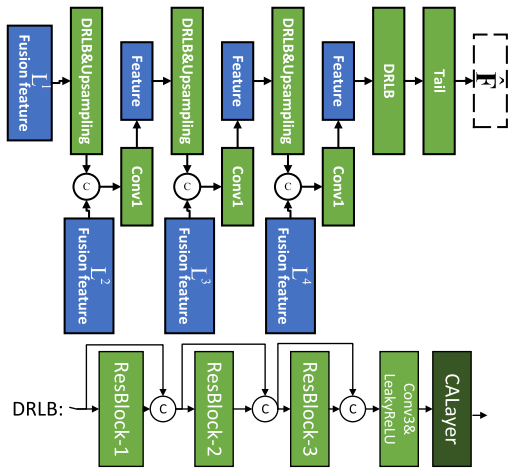


Fig. 10. Overall network of multilevel reconstruction.

The DRLB and upsampling block is the key block here, which consists of two parts: the DRLB and upsampling. The DRLB is generated according to DRLN [45], which is composed of densely connected Resblock, a channel-sort block (named Conv3&LeakyReLU), and Laplacian channel attention (named CALayer in Fig. 10). The three Resblocks that are set in a row are used for feature extraction. As the convolutional layer deepens, the receptive field also expands, and the features of different depths are obtained. Then, we use a short connection to constantly reconcatenate the input before the next Resblock, which aims to prevent the loss of information from the deep network and get the characteristics of different receptive fields in the output. The channel-sort block uses a convolution with a kernel size of 3 and a channel number of 64 and uses LeakyReLU as the activation function. The Laplacian channel attention uses the global average pooling and the dilated convolution of different convolutions to generate features of different scales. The combination of different scales is then passed through a convolution and sigmoid function to assign weights to each channel, which obtains the channel attention.

The specific convolution parameters for the fusion module are presented in Table II.

E. Discriminator

Here, we utilize the U-net [47] network structure as our discriminator because U-nets have demonstrated state-of-art performance in many complex image segmentation tasks [48], [49]. Since it has the ability to process global and local information at the same time, it will be a satisfying network for the discriminator in our spatial fusion task. The detailed architecture of our U-net-based discriminator is shown in Fig. 11. It contains two parts: the downsampling encoder and the upsampling decoder network. The downsampling encoder continuously downsamples the image to capture global information, while the decoder uses progressive upsampling to match the input and output resolutions. Meanwhile, the decoder is connected by skipping connections to further improve the processing capability of details. Overall speaking, the numbers of channels for each DownsampleBlock in the downsampling encoder are 16, 32, and 64, respectively, and the numbers in the upsampling decoder are 64, 32, and 16. The specific parameters of some key blocks in the dashed box are tabulated in Table III. Finally, we use a SigmoidBlock for judgment. The final output is a 2-D matrix with the same size as the input image, which indicates that the possibility of the input image is true. In the training step of the discriminator, if the input is $\widehat{\mathbf{F}}$, we expect the output to be a full false matrix, and if the input is $\mathbf{F}$, we expect the output is a full true matrix.

TABLE II
PARAMETERS OF BLOCKS IN THE FUSION MODULE

	Kenel size	Stride	Number of channels
Head	1	1	64
Downsampling convolution	3	2	64
Local attention-Conv1	1	1	64
Global attention-Conv1	3	1	64
Upsampling deconvolution	3	2	64
DRLB-resblock-1	3	1	64
DRLB-resblock-2	3	1	128
DRLB-resblock-3	3	1	256
DRLB-conv3	3	1	64
Tail	1	1	3

TABLE III
PARAMETERS OF BLOCKS IN DASHED BOXES OF FIG. 11

	Kenel size	Stride	Channel number
Conv3	3	1	64
Downsample convolution	3	2	64
Upsample deconvolution	3	2	64

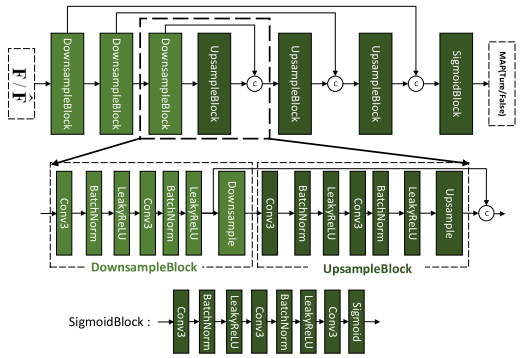


Fig. 11. Overall network of discriminator. The downsampling encoder blocks are colored in light green, while the upsampling decoder ones are in dark green.

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F. Loss Function

The commonly used LSGAN loss ($\mathcal{L}_{\mathrm{LSGAN}}$) is adopted here, together with additional constraints on the image content. Compared with the traditional GAN, LSGAN is designed to keep the data distribution from deviating too much, making the training more stable. The expression for $\mathcal{L}_{\mathrm{LSGAN}}$ is

$$\mathcal{L}_{\mathrm{LSGAN}}(D) = \frac{1}{2}\mathbb{E}_{\mathbf{F}}\left((D(\mathbf{F})-1)^2\right) + \frac{1}{2}\mathbb{E}_{\mathbf{C}_1,\mathbf{C}_2,\mathbf{F}_{\mathrm{SSM}}}\left(D(G(\mathbf{C}_1,\mathbf{C}_2,\mathbf{F}_{\mathrm{SSM}}))^2\right) \quad (6)$$

$$\mathcal{L}_{\mathrm{LSGAN}}(G) = \frac{1}{2}\mathbb{E}_{\mathbf{C}_1,\mathbf{C}_2,\mathbf{F}_{\mathrm{SSM}}}\left((D(G(\mathbf{C}_1,\mathbf{C}_2,\mathbf{F}_{\mathrm{SSM}}))-1)^2\right) \quad (7)$$

where $\mathbb{E}(.)$ denotes the expectation function. $\mathcal{L}_{\mathrm{LSGAN}}(D)$ is the final loss function for the discriminator. $\mathcal{L}_{\mathrm{LSGAN}}(G)$ is used as a part of the loss for the generator. The losses for the image itself mainly include content loss and structure loss. For the content loss, the $\mathcal{L}_1$ loss instead of $\mathcal{L}_2$ loss is adopted because using the latter one will make the predicted image too smooth

$$\mathcal{L}_1 = \frac{1}{N}\sum_{i=1}^{N}\|\mathbf{T}-\mathbf{P}\|_1 \quad (8)$$

where $\mathbf{T}$ is the true image at the predicted moment and $\mathbf{P}$ is the predicted image. $\|\cdot\|_n$ is the n-norm, and N is the total number of pixels. The multiscale SSIM (MS-SSIM) [50], which calculates the multiscale similarity of $\mathbf{T}$ and $\mathbf{P}$, is adopted to compute the structure loss

$$\mathcal{L}_{\mathrm{structure}} = 1 - \text{MS-SSIM}(\mathbf{T},\mathbf{P}) \quad (9)$$

where MS-SSIM(.) represents the MS-SSIM function [50].

Then, the total generator loss function of SRSF-GAN is

$$\mathcal{L}(G) = \alpha\mathcal{L}_{\mathrm{GAN}}(G) + \beta\mathcal{L}_1\left(\widehat{\mathbf{F}},\mathbf{F}\right) + \lambda\mathcal{L}_{\mathrm{structure}}\left(\widehat{\mathbf{F}},\mathbf{F}\right) + \xi\mathcal{L}_{\mathrm{structure}}\left(\widehat{\mathbf{F}}_{\mathrm{SR}},\mathbf{F}\right) \quad (10)$$

where α, β, λ, and ξ are the weight coefficients. It should be noted here that the structure loss of the predicted SR image ($\widehat{\mathbf{F}}_{\mathrm{SR}}$) and true image ($\mathbf{F}$) is also included here. It is because the reconstruction information for changed areas is mainly from $\mathbf{F}_{\mathrm{SR}}$. By doing so, it can enhance the ability of our network to predict images with great changes.

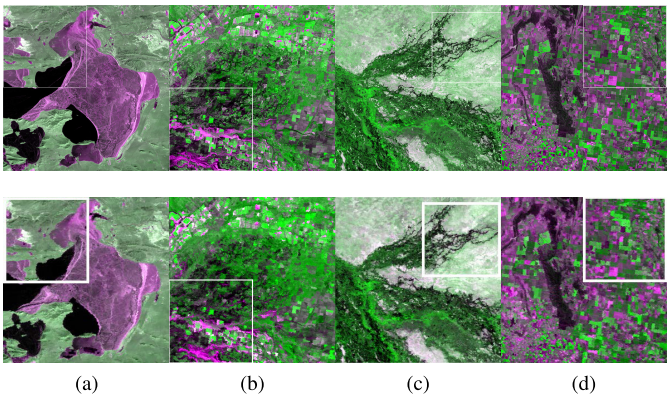
III. Experiments and Results

A. Study Areas and Datasets

Currently, many spatiotemporal fusion models are developed or trained on ideal datasets, neglecting the difficulties in data preparation [27]. Thereby, in this article, we build four datasets with ground objects changed with different degrees in spatial and/or spectral dimension, as shown in Fig. 12.

1) *Lake Data:* The first study area is located at the Selincuo Lake (latitude: 32.1488°N, longitude: 88.6592°E) on the Qinghai–Tibet Plateau, China, covering an area of 172.8 km^2. During the freezing (melting) season, the land cover type of the lake can be partly or totally changed from water to ice (ice to water). It is used to test the performance of our method when the ground objects have a significant spatial change. The dataset covers 16 pairs of Landsat-MODIS images from 2015 to 2020. The Landsat images are from the Landsat 8 surface reflectance product (https://earthexplorer.usgs.gov/), which contains seven bands from visible to short-wave infrared spectral ranges, while the MODIS images are MOD09GQ Collection 6 (https://lpdaac.usgs.gov/products/mod09gqv006/), which have only two bands (the red and near-infrared bands). Thus, only the red and near-infrared bands of Landsat data are used here. Besides, both the spatial resolution and map projection of the two images are different: the Landsat 8 image has a spatial resolution of 30 m in the UTM projection, while the MOD09GQ image has a spatial resolution of 250 m in the sinusoidal projection. In order to unify the projection and also satisfy the 2× upsampling/downsampling scale, we convert all the MOD09GQ images to the UTM projection with a spatial resolution of 240 m. The size of the Landsat image is 240 × 240, and the size of the MODIS image is 30 × 30. The MOD09GQ images of the left two datasets have been processed in the same way, and the sizes of both Landsat and MODIS images of the following two datasets are the same.
2) *Flood Data:* The second dataset is also located in southern Australia (latitude: 29.0855°S, longitude: 149.2815°E). This area is relatively homogeneous with large parcels of croplands and natural vegetation. However, a large flood occurred in December 2004. As a result, the original farmland/natural vegetation areas were covered by water, leading to significant changes in land cover types. The main purpose of this data is to test the predictive ability of the algorithms under the condition of complex land cover changes. The dataset covers 13 pairs of Landsat-MODIS images from 2004 to 2005.
3) *Delta Data:* The third study area with a complex wetland landscape is at the Okavango Delta (latitude: 18.6081°S, longitude: 22.4436°E) in northwestern Botswana with an area of 172.8 km^2. Due to the seasonal flooding, the water volume of the meandering rivers in the delta changes accordingly. This dataset is chosen to evaluate the ability of our method to reconstruct the abrupt change with a size smaller than the spatial resolution of the coarse image. The dataset covers 21 pairs of Landsat-MODIS images from 2015 to 2020.
4) *Farmland Data:* The last dataset is located in the irrigated farmland in southern Australia (latitude: 33.7344°S, longitude: 145.9721°E), named Farmland data. The land cover and land use types of this area are temporally stable, yet the spectral characteristics may change due to vegetation growth and phenology change. Moreover, this site is relatively homogeneous with lots of large-sized crop fields and natural vegetation. The main purpose of using this dataset is to test the performance of our method in monitoring seasonal changes in fragmented cropland landscapes. In addition, we also largely reduce the number of training data compared to the above two datasets. That is, this dataset only covers

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(a) (b) (c) (d)

Fig. 12. False color image (R: red; G: near-infrared; and B: red) of the Landsat (the upper, 30 m) and MODIS (lower, 240 m) data for the four study areas, which are (a) lake data (January 18, 2018), (b) flood data (December 12, 2004), (c) delta data (June 12, 2020), and (d) farmland data (October 13, 2020). Areas indicated by a white rectangular box are the prediction ones.

six pairs of Landsat-MODIS images from 2018 to 2020. Therefore, this dataset is also used to test the performance of our method when the training data are limited. It should be noted here that the study areas of the flood and farmland data are both consistent with the two commonly used standard datasets in the Coleambally Irrigation Area (CIA) study site and the Lower Gwydir Catchment (LGC) study site. Yet, since our method requires the 250-m MOD09GQ data as input, we redownload the coarse-resolution data in this article.

B. Experiment Design

In this section, we totally conduct four series of experiments to validate the performance of our method.

1) *Comparison Experiments:* Quantitative comparison experiments are conducted with state-of-the-art algorithms.
2) *Ablation Study:* Ablation experiments of the GAN structure, SR module, and Fusionblock.
3) *Data Selection Study:* Two series of experiments to show the influence of data selection about the coarse-resolution images on the performance of our method.
4) *Computational Load Comparison:* Compare model parameter quantities of our method to state-of-the-art algorithms.

For the comparison experiments, we first conduct a parameter setting experiment to determine the best parameters for our method. Then, we compare our method with six state-of-the-art algorithms, namely, STARFM [8], FSDAF [15], EDSTFN [26], GANSTFM [27], MLFF-GAN [29], and SWINSTFM [34] on the four study areas mentioned earlier. For quantitatively comparison, four commonly used accuracy metrics, i.e., root mean square error (RMSE), mean absolute error (MAE), SSIM [38], and universal image quality index (UIQI) [51], are used here to measure the consistency between the predicted image and the ground truth.

It should be noted that all six comparison methods belong to the category of single-phase reference image pattern. For this pattern, the input reference images should be selected based on the temporal interval criterion. In addition, our method uses the SSM criterion for the reference image selection. It can be expected that different reference images will be selected if the criterion is not unified. As discussed in Section II-A, the SSM metric is more reasonable than the temporal interval. Therefore, we use the aSSIM to select the best Landsat image as a reference. For a fair comparison, all the methods use this fine image and the corresponding coarse image (if necessary) as the reference data.

The former two methods (i.e., STARFM and FSDAF) are non-DL-based. Meanwhile, the left methods are all DL-based. Except for EDSTFN and SWINSTFM, all the DL methods, including our model, use the Adam optimizer with a learning rate of 0.0002 in the generator and a learning rate of 0.002 in the discriminator. For EDSTFN and SWINSTFM, the learning rate is set to 0.0002. All DL models are trained for 100 epochs on the lake and farmland data but for 160 epochs on the delta and flood data to avoid underfitting. In addition, since all six comparison methods require the input coarse images to have the same size as the fine image, we upsample all coarse images by bilinear interpolation. In addition, the size of the Landsat and MODIS images used in SWINSTFM is 256×256, while the size of the images used in the other five methods is 240×240.

As for the ablation study, first, we remove the structure of GAN, only keep the part of the generator, and disable the adversarial loss, so as to prove that the adversarial network has an enhancement effect on our model. Then, we ablate two important modules in the model, i.e., the SR and fusion modules. Since our SR module is controlled by the SR loss function, we remove the SR loss function as the ablation of this module in the experiment. On the other hand, because the fusion attention mechanism (namely, Fusionblock) is more important, we directly add the reference image features and the coarse-resolution image features at the predicted time without going through our attention mechanism to achieve the ablation of the fusion module.

C. Comparison Experiments

1) Parameter Setting: As presented in (10), there are four losses in our loss functions. Among them, the first ($\mathcal{L}_{\text{GAN}}(G)$) and last loss ($\mathcal{L}_{\text{structure}}(\widehat{\mathbf{F}}, \mathbf{F})$) are essential. $\mathcal{L}_{\text{GAN}}(G)$ is the GAN loss, which guides the generation of high-quality images, while $\mathcal{L}_{\text{structure}}(\widehat{\mathbf{F}}, \mathbf{F})$ measures the difference between the final fusion image and the ground-truth image. To determine the best parameter settings, we conduct parameter experiments with $\alpha = [0.01, 0.001]$ and $\xi = [0.25, 0.5, 0.75, 1]$ on the four datasets. For the left two parameters (β, λ), we fix them to 1. Therefore, there are 32 experiments in total. The results are shown in Table IV. As can be seen from the table, the performance of the model remains relatively stable with different weighted loss functions. Even in the worst case scenario, our algorithm can still maintain excellent results in comparative experiments. For our model, we set the weights in the loss function as $\alpha = 0.01$, $\beta = \lambda = 1$, and $\xi = 0.75$ for delta data and $\alpha = 0.01$, $\beta = 1$, $\lambda = 1$, and $\xi = 1$ for the all other datasets.

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TABLE IV

EXPERIMENTAL RESULTS OF LOSS FUNCTION PARAMETERS FOR DATASETS. NUMBERS IN BOLD INDICATE THE BEST VALUES

dataset	α	ξ	RMSE	MAE	SSIM	UIQI
Lake Data	0.01	0.25	0.0706	0.0525	0.7728	0.9468
		0.5	0.0634	0.0442	0.7521	0.9516
		0.75	0.0601	0.0392	0.7712	0.9564
		1	**0.0555**	0.0332	**0.7872**	**0.9605**
	0.001	0.25	0.0578	0.0358	0.7820	0.9568
		0.5	0.0564	**0.0327**	0.7848	0.9584
		0.75	0.0606	0.0377	0.7463	0.9551
		1	0.0569	0.0331	0.7840	0.9594
Delta Data	0.01	0.25	0.0564	0.0386	0.8759	0.9669
		0.5	0.0589	0.0419	0.8757	0.9654
		0.75	**0.0491**	**0.0296**	0.8853	**0.9726**
		1	0.0580	0.0433	**0.8855**	0.9684
	0.001	0.25	0.0519	0.0347	0.8822	0.9704
		0.5	0.0491	0.0298	0.8779	0.9725
		0.75	0.0572	0.0410	0.8746	0.9696
		1	0.0672	0.0532	0.8770	0.9614
Farmland Data	0.01	0.25	0.1256	0.0934	**0.6369**	0.7770
		0.5	0.1420	0.1116	0.5834	0.7581
		0.75	0.1408	0.1077	0.6045	0.7569
		1	**0.1126**	**0.0843**	0.6231	**0.8265**
	0.001	0.25	0.1343	0.1110	0.5185	0.7936
		0.5	0.1747	0.1316	0.6095	0.6707
		0.75	0.1432	0.1081	0.6208	0.7323
		1	0.1768	0.1296	0.6093	0.6381
Flood Data	0.01	0.25	0.1044	0.0754	0.4960	0.8332
		0.5	0.1186	0.0875	0.4886	0.7899
		0.75	0.1467	0.1139	0.4860	0.7779
		1	**0.1001**	**0.0725**	0.5117	**0.8414**
	0.001	0.25	0.1030	0.0748	**0.5184**	0.8202
		0.5	0.1026	0.0734	0.4983	0.8220
		0.75	0.1046	0.0744	0.5084	0.8171
		1	0.1166	0.0868	0.4792	0.7969

2) Lake Data: Fig. 13 presents the Landsat-like image for part of Selincuo Lake on January 18, 2018, predicted by the compared five methods and our model. A zoomed-in area is also distributed aside to highlight the difference between predicted images and the ground truth. The error maps are further calculated to present the spatial distribution of the difference in Fig. 14. Comparing Fig. 13(a) and (b), we can see that the main differences include the following areas.

1) *Ice-to-Water Changed Areas:* Area A, Area B, Area C, and Area F.
2) *Water-to-Ice Changed Areas:* Area D and Area E.

Among them, the size of Areas A and B is larger than the MODIS's spatial resolution (especially Area B), the size of Areas B and E is comparable to the MODIS's spatial resolution, and the size of Area F is smaller than the MODIS's spatial resolution.

Generally, all the DL methods (i.e., EDCSTFN, GANSTFM, MLFF-GAN, SWINSTFM, and SRSF-GAN) can achieve better performance than the non-DL ones (i.e., STARFM and FSDAF). For example, for the large-sized changed areas, such as Areas A and B, all the methods can predict the water pixels correctly. Yet, for the medium-sized area, i.e., Area D, only DL methods can predict the correct ice pixels. Moreover, FSDAF is not able to reconstruct the ice area at Area E. In addition, from Fig. 13(c) and (d), we can see that both STARFM and FSDAF have generated a lot of noise on land, water, and ice regions. From the error maps (see Fig. 14), it can also be found that the average error between the ground truth and the predicted images by the non-DL methods is larger than that by the DL methods.

TABLE V

ACCURACY METRICS OF STARFM, FSDAF, EDCSTFN, GANSTFM, MLFF-GAN, SWINSTFM, AND SRSF-GAN FOR THE LAKE DATA. NUMBERS IN BOLD INDICATE THE BEST VALUES, WHILE NUMBERS WITH THE UNDERLINE INDICATE THE SECOND GOOD VALUES IN EACH COLUMN

Model	RMSE	MAE	SSIM	UIQI
STARFM	0.2651	0.2256	0.5115	0.4636
FSDAF	0.2672	0.2252	0.4680	0.4491
EDCSTFN	0.0770	0.0532	0.7014	0.9140
GANSTFM	<u>0.0724</u>	<u>0.0472</u>	<u>0.7059</u>	<u>0.9251</u>
MLFF-GAN	0.0841	0.0599	0.6902	0.8988
SWINSTFM	0.2114	0.1753	0.6512	0.7824
SRSF-GAN	**0.0555**	**0.0332**	**0.7872**	**0.9605**

As for the DL methods, both EDCSTFN and GANSTFM will mistakenly predict some ice at the edge of Area B [see Fig. 13(e) and (f)]. The output image of SWINSTFM [see Fig. 13(h)] is too bright, especially for the ice area. Finally, all the other DL methods are not able to reconstruct small-sized melting/freezing areas, which could be water or covered by thin ice at Area C. Yet, our method can successfully predict most of those areas with abundant details. In terms of the error map, our model has significantly lower values than the other three DL algorithms. However, none of the six methods can regenerate the narrow thin ice lines at Area F. It is because both the coarse images (due to the resolution limitation) and the Landsat reference images cannot provide this prior knowledge. Though the SR module in SRSF-GAN has the ability to predict changed areas according to the coarse image at t_1, it will fail when the size of the changed area is much smaller than the resolution of the coarse image.

The quantitative evaluation of the six methods on the Lake data is tabulated in Table V. It can be seen that our method can derive the lowest RMSE and MAE, indicating that the predicted image of our method is closest to the ground truth from the perspective of data value. Moreover, SRSF-GAN can also obtain the highest values for SSIM and UIQI, which further infers that our method can reconstruct more details than the other methods. In all, SRSF-GAN can achieve the best performance on the Lake data.

3) Flood Data: Due to the suddenness of the flood, it is quite possible that there exist large differences between the reference high-resolution image and the ground truth of the predicted target. As a result, it will cause a great challenge to algorithms' prediction ability. Comparing Fig. 15(a) and (b), we mainly focus on the following three areas.

1) *Vegetation Areas With Minor Changes:* Area A and Area B.
2) *Dry Soil-to-Water (Flooding Areas):* Area C.
3) *Dry Soil-to-Wet Soil:* Area D.

Upon visual analysis, we can find that traditional methods (i.e., STARFM and FSDAF) overpredict the flooding area (Area C) and are unable to predict the wet soil area (Area D). Moreover, they cannot always obtain a good performance on the vegetation areas with small changes (Areas A and B). Generally, DL methods can perform well on vegetation areas with small changes, especially in Area A. For Area C,

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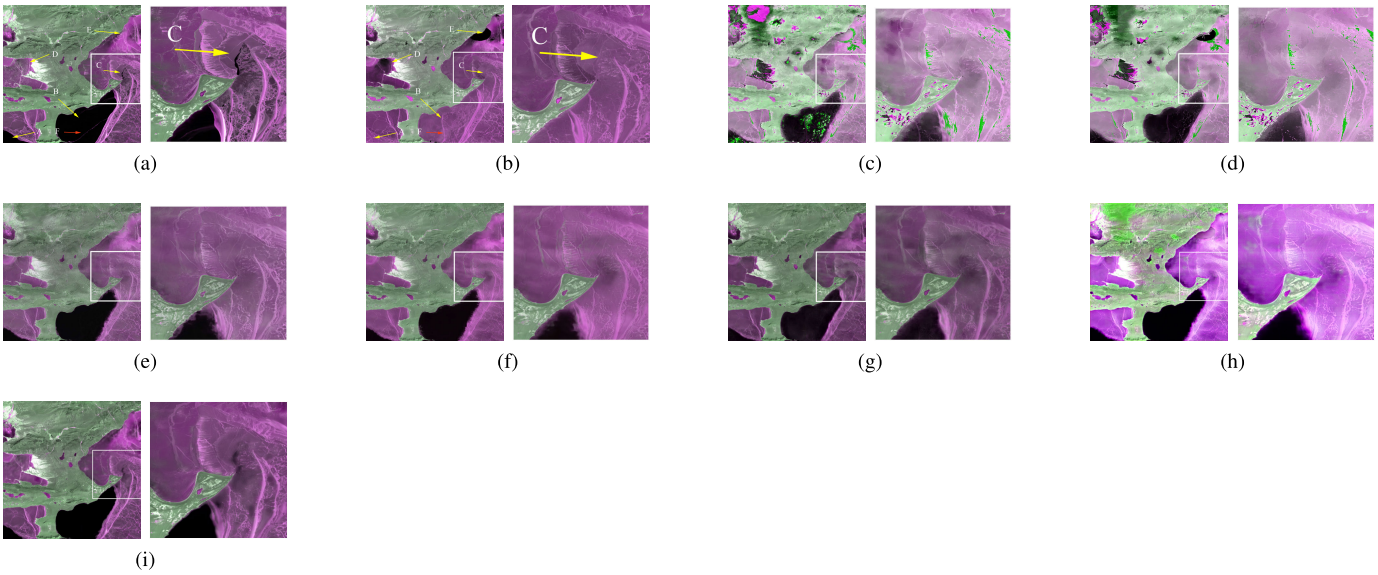


Fig. 13. Fusion results for the lake data on January 18, 2018. (a) Ground truth. (b) Reference fine image on March 7, 2018. (c) STARFM. (d) FSDAF. (e) EDCSTFN. (f) GANSTFM. (g) MLFF-GAN. (h) SWINSTFM. (i) SRSF-GAN. In each group, the right image is an enlarged one corresponding to the rectangular box of the predicted image on the left.

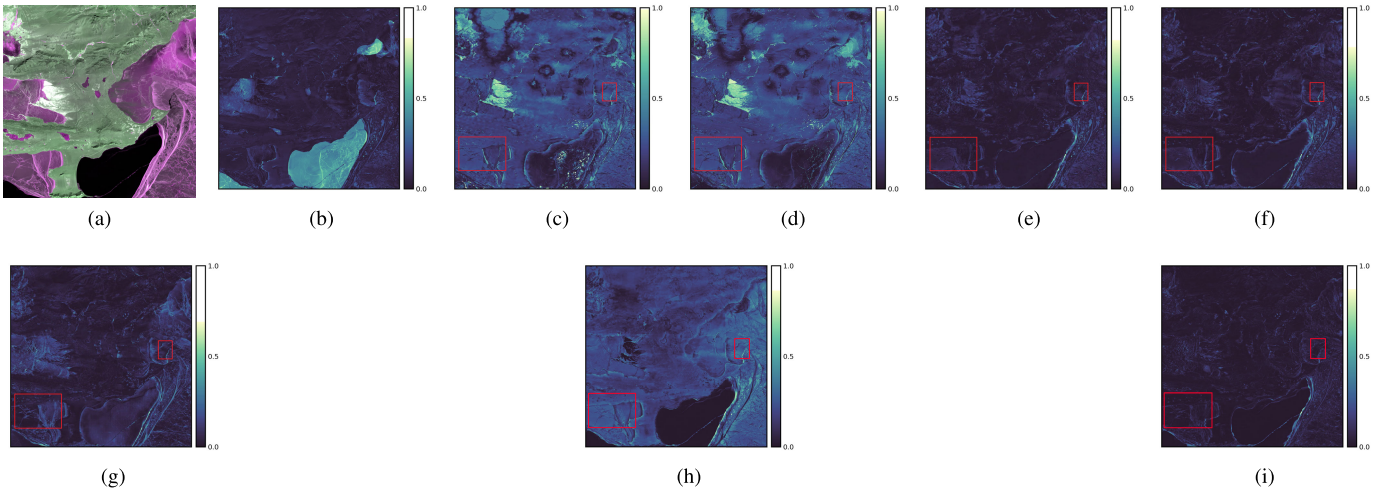


Fig. 14. (a) Ground truth. The error maps of (b) reference fine image, (c) STARFM, (d) FSDAF, (e) EDCSTFN, (f) GANSTFM, (g) MLFF-GAN, (h) SWINSTFM, and (i) SRSF-GAN for the lake data.

SWINSTFM also overpredicts the flooding area. Moreover, its result is brighter, while EDCSTFN's and MLFF-GAN's results are darker in this flooding area. For Area D, GANSTFM can only predict a small part of this area. The other DL algorithms can generally generate more wet soil pixels, yet they are not able to predict the edges very well. Overall, our method can achieve the best result visually. The error maps are shown in Fig. 16; one can find that the traditional algorithms have a larger error than the DL methods, especially STARFM. The quantitative comparison of the four accuracy metrics is tabulated in Table VI. We can see that our algorithm can achieve the best performance on this dataset, and GANSTFM has a suboptimal performance.

TABLE VI

ACCURACY METRICS OF STARFM, FSDAF, EDCSTFN, GANSTFM, MLFF-GAN, SWINSTFM, AND SRSF-GAN FOR THE FLOOD DATA. NUMBERS IN BOLD INDICATE THE BEST VALUES, WHILE NUMBERS WITH THE UNDERLINE INDICATE THE SECOND GOOD VALUES IN EACH COLUMN

Model	RMSE	MAE	SSIM	UIQI
STARFM	0.1685	0.1184	0.3920	0.6188
FSDAF	0.1334	0.1045	0.4322	0.7568
EDCSTFN	0.1189	0.0871	0.4981	0.7293
GANSTFM	<u>0.1031</u>	<u>0.0737</u>	<u>0.5035</u>	<u>0.8297</u>
MLFF-GAN	0.1189	0.0871	0.4981	0.7293
SWINSTFM	0.1389	0.1001	0.4914	0.7949
SRSF-GAN	**0.1001**	**0.0725**	**0.5117**	**0.8414**

4) Delta Data: Comparing Fig. 17(a) and (b), we can find that the two meandering rivers (marked as Areas A and B) on the prediction date are changed to dry lands on the fine reference image. Moreover, the size of rivers (about 90–180 m) is much smaller than the spatial resolution of MODIS (i.e., 250 m). Thereby, recovering the two rivers with details is of great challenge for all fusion methods.

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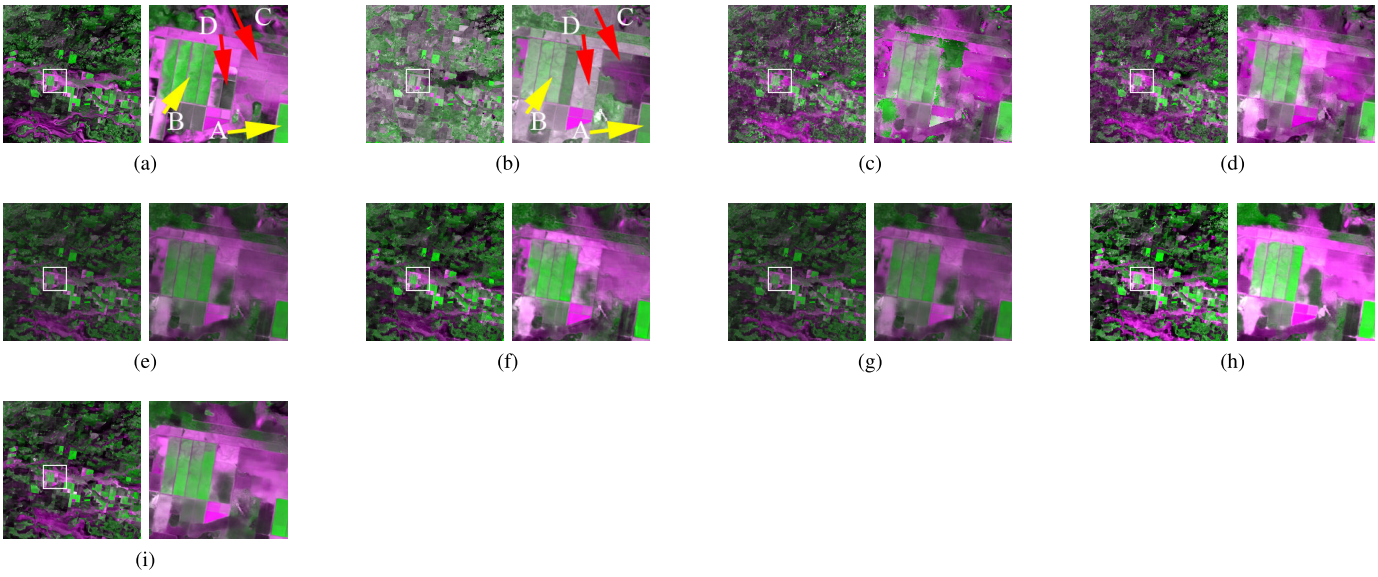


Fig. 15. Fusion results for the flood data on December 12, 2004. (a) Ground truth. (b) Reference fine image on November 26, 2004. (c) STARFM. (d) FSDAF. (e) EDCSTFN. (f) GANSTFM. (g) MLFF-GAN. (h) SWINSTFM. (i) SRSF-GAN. In each group, the right image is an enlarged one corresponding to the rectangular box of the predicted image on the left.

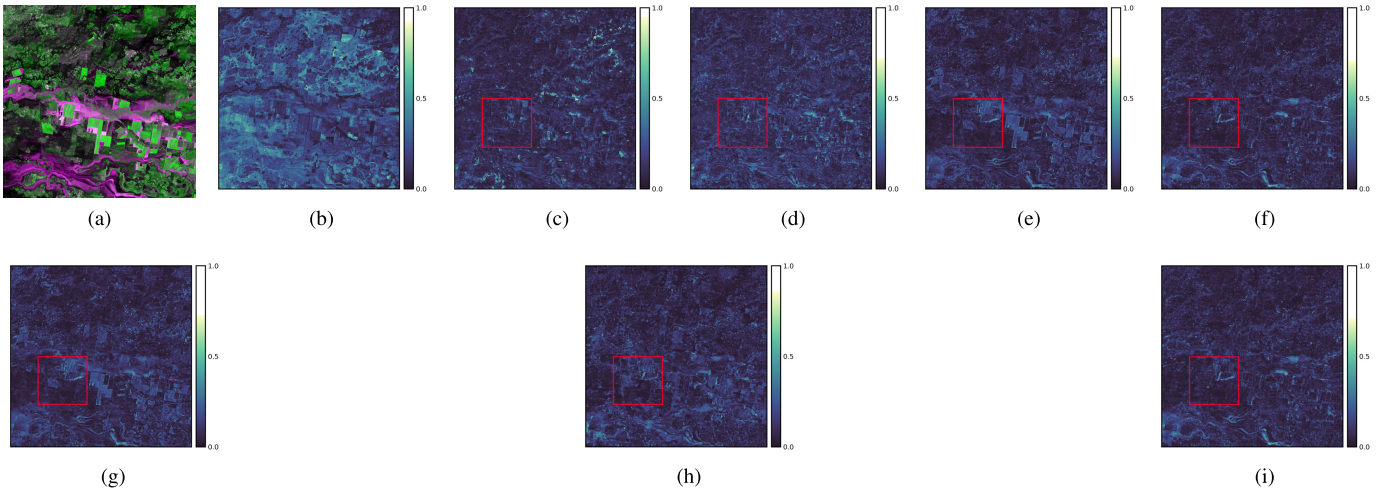


Fig. 16. (a) Ground truth. The error maps of (b) reference fine image, (c) STARFM, (d) FSDAF, (e) EDCSTFN, (f) GANSTFM, (g) MLFF-GAN, (h) SWINSTFM, and (i) SRSF-GAN for the flood data.

The predicted results are demonstrated in Fig. 17, and the error maps are shown in Fig. 18. Like the results on the Lake data, the non-DL methods still produce noisy pixels on the Delta data. Moreover, both STARFM and FSDAF can hardly recover the two rivers, not to mention reconstructing the details. Comparatively, the DL methods can achieve a better result on this dataset. Especially, MLFF-GAN and SRSF-GAN can mostly reconstruct the two rivers. Yet, when examining the enlarged images, it can be seen that our method can predict more details. Moreover, SRSF-GAN also outperforms MLFF-GAN from the perspective of the error map. The quantitative comparison results tabulated in Table VII also prove that SRSF-GAN can achieve the best performance on the Delta data. It infers that SRSF-GAN has the ability to predict small-sized changed areas with satisfactory accuracy. Moreover, one can also find that, even with less input information (i.e., the coarse image corresponding to the reference image), SRSF-GAN can still make more accurate results than DL methods in the single-phase reference image pattern. The main reason is that SRSF-GAN involves the SR module in the fusion progress to improve the ability to predict changed areas and details.

5) Farmland Data: Unlike the above three datasets, there are no land cover and land use type changes in farmland data, but only the spectral variance exists due to the farmland growth and harvest. In addition, this site is relatively homogeneous with large-sized fields. As discussed in previous studies [8], [15], some non-DL methods can solve this kind of fusion problem with a satisfactory result. As shown in Figs. 19 and 20, FSDAF is able to produce a good prediction with better color contrast and a lower error map. What is more, from Table VIII, we can also see that FSDAF can get the largest UIQI value. Yet, the other non-DL method, STARFM, cannot make a comparable result as FSDAF. Visually, STARFM has made a lot

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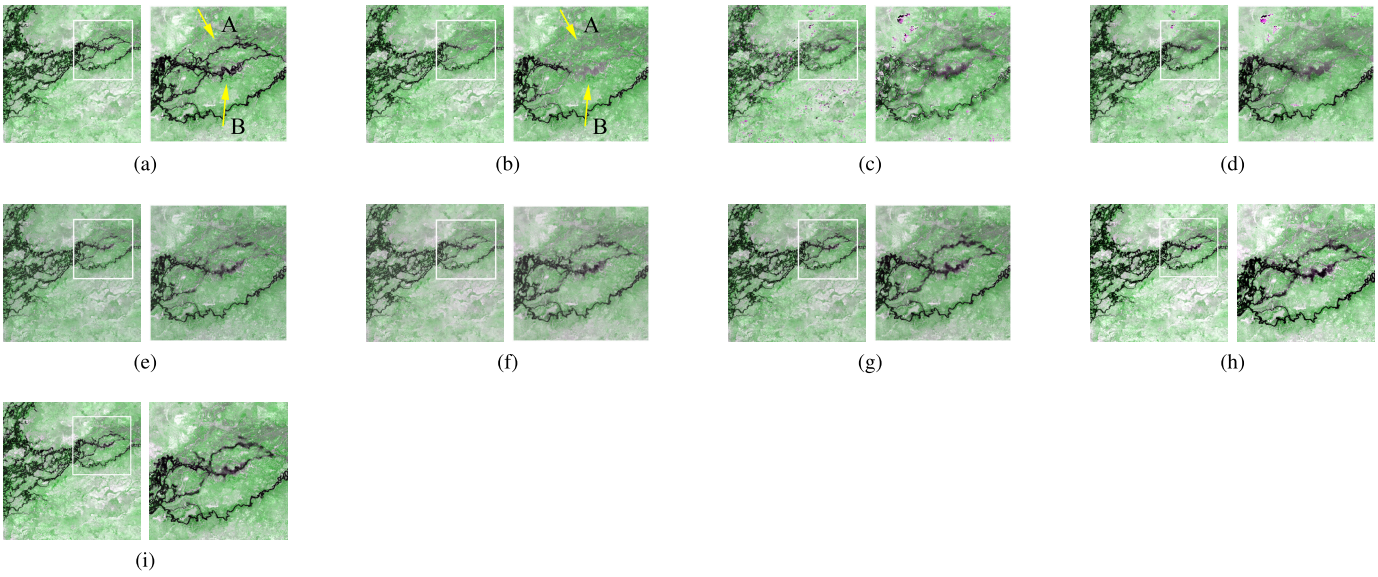


Fig. 17. Fusion results for the delta data on June 12, 2020. (a) Ground truth. (b) Reference image for May 27, 2020. (c) STARFM. (d) FSDAF. (e) EDCSTFN. (f) GANSTFM. (g) MLFF-GAN. (h) SWINSTFM. (i) SRSF-GAN. In each group, the right image is an enlarged one corresponding to the rectangular box of the predicted image on the left.

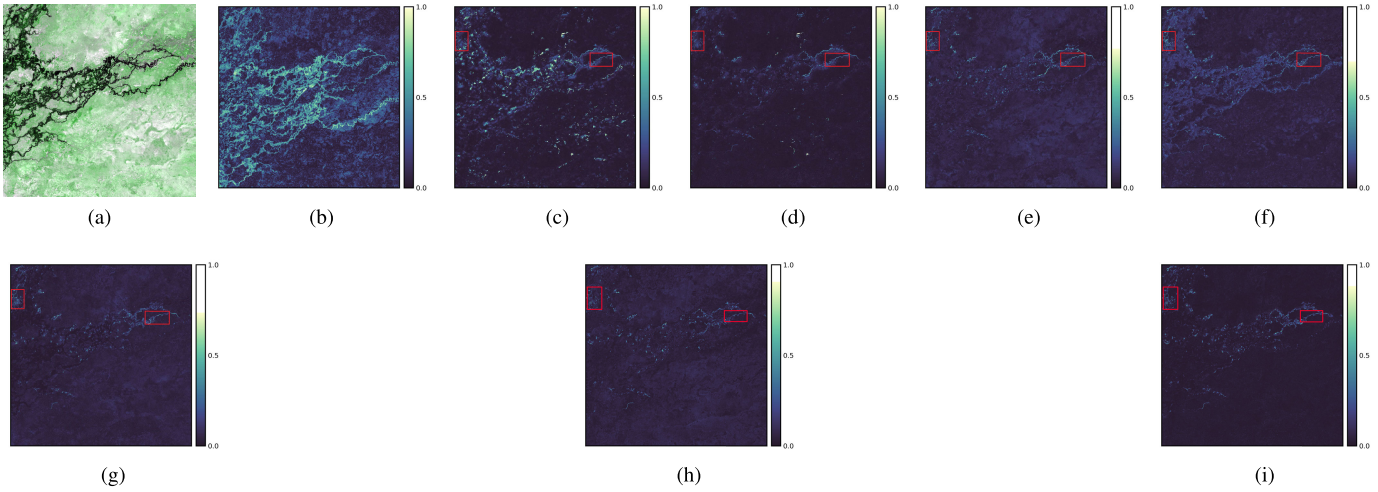


Fig. 18. (a) Ground truth. The error maps of (b) reference fine image, (c) STARFM, (d) FSDAF, (e) EDCSTFN, (f) GANSTFM, (g) MLFF-GAN, (h) SWINSTFM, and (i) SRSF-GAN for the delta data.

TABLE VII

ACCURACY METRICS OF STARFM, FSDAF, EDCSTFN, GANSTFM, MLFF-GAN, SWINSTFM, AND SRSF-GAN FOR THE DELTA DATA. NUMBERS IN BOLD INDICATE THE BEST VALUES, WHILE NUMBERS WITH THE UNDERLINE INDICATE THE SECOND GOOD VALUES IN EACH COLUMN

Model	RMSE	MAE	SSIM	UIQI
STARFM	0.1236	0.0602	0.7243	0.8141
FSDAF	0.0675	0.0379	0.8303	0.9476
EDCSTFN	0.0780	0.0652	0.8601	0.9327
GANSTFM	0.0760	0.0586	0.8026	0.9287
MLFF-GAN	0.0629	0.0504	0.8783	0.9601
SWINSTFM	0.0742	0.0589	0.8708	0.9630
SRSF-GAN	**0.0491**	**0.0296**	**0.8853**	**0.9726**

TABLE VIII

ACCURACY METRICS OF STARFM, FSDAF, EDCSTFN, GANSTFM, MLFF-GAN, SWINSTFM, AND SRSF-GAN FOR THE FARMLAND DATA. NUMBERS IN BOLD INDICATE THE BEST VALUES, WHILE NUMBERS WITH THE UNDERLINE INDICATE THE SECOND GOOD VALUES IN EACH COLUMN

Model	RMSE	MAE	SSIM	UIQI
STARFM	0.1728	0.1174	0.4907	0.6211
FSDAF	0.1277	0.0947	0.5387	0.7919
EDCSTFN	0.1498	0.1211	0.6023	0.7346
GANSTFM	0.1195	0.0892	0.5199	0.6826
MLFF-GAN	0.1578	0.1354	0.5974	0.7060
SWINSTFM	0.1931	0.1464	0.4794	0.6858
SRSF-GAN	**0.1126**	**0.0843**	**0.6231**	**0.8265**

of errors in the prediction results [see Figs. 19(c) and 20(c)]. Correspondingly, it has the largest RMSE and the lowest SSIM and UIQI (see Table VIII), indicating that it performs worst on this dataset.

Visually, the results generated by the DL methods are not superior to that by FSDAF. For example, the overall color of the image generated by GANSTFM is darker than the ground truth, while the color of the image by MLFF-GAN is brighter.

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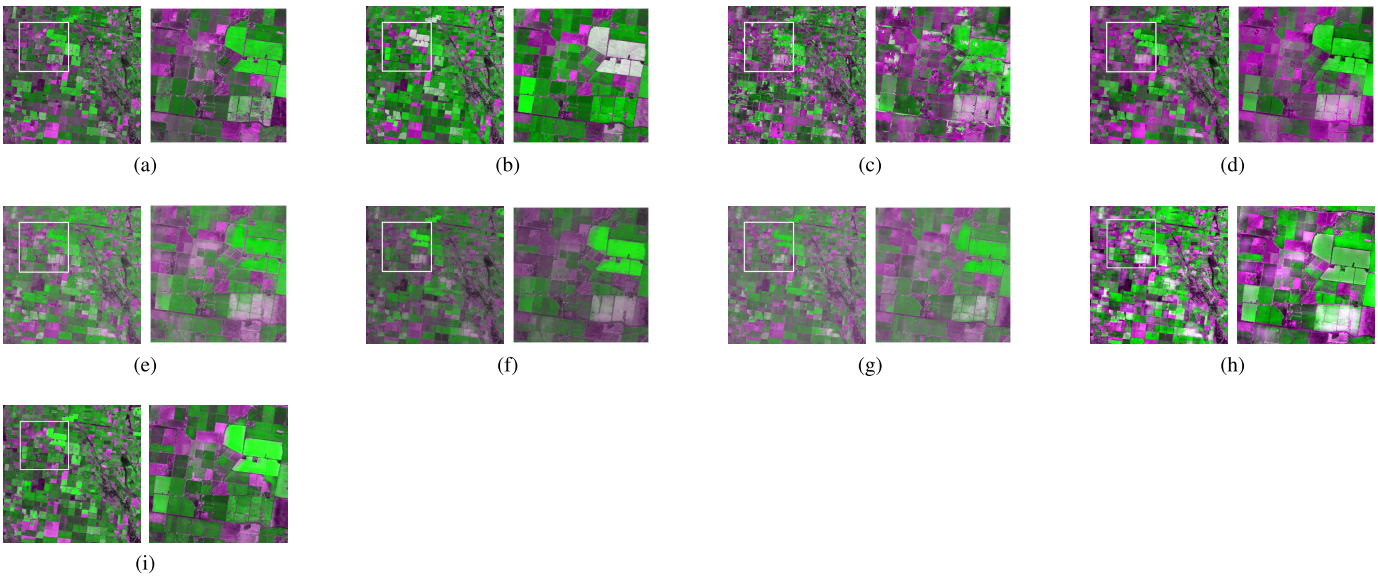


Fig. 19. Fusion results for the farmland data on October 13, 2020. (a) Ground truth. (b) Reference image for the farmland data on September 11, 2020. (c) STARFM. (d) FSDAF. (e) EDCSTFN. (f) GANSTFM. (g) MLFF-GAN. (h) SWINSTFM. (i) SRSF-GAN. In each group, the right image is an enlarged one corresponding to the rectangular box of the predicted image on the left.

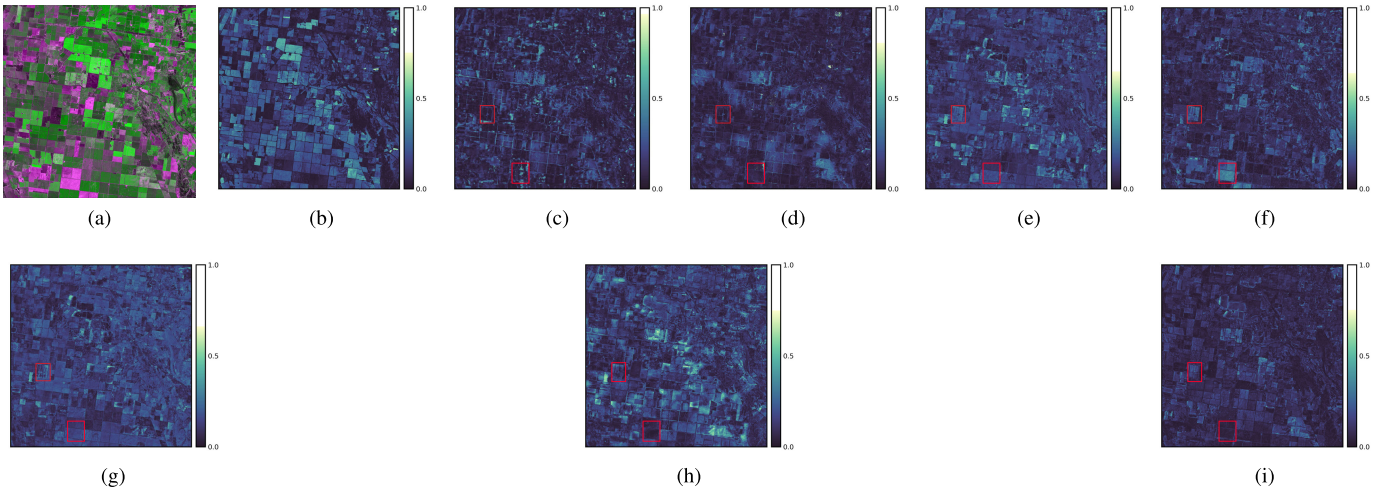


Fig. 20. (a) Ground truth. The error maps of (b) reference fine image, (c) STARFM, (d) FSDAF, (e) EDCSTFN, (f) GANSTFM, (g) MLFF-GAN, (h) SWINSTFM, and (i) SRSF-GAN for the farmland data.

This is mainly because the training data of this study area are not enough. However, the quantitative comparison shows that our method can still achieve the lowest RMSE and MAE and the highest SSIM and UIQI (see Table VIII). It means that, even with limited training data, SRSF-GAN can still perform better than the well-performed traditional fusion method (such as FSDAF).

TABLE IX
ABLATION EXPERIMENT OF THE GAN STRUCTURE

	Dataset	RMSE	MAE	SSIM	UIQI
With GAN	Lake	**0.0555**	**0.0332**	**0.7872**	**0.9605**
	Flood	**0.1001**	**0.0725**	**0.5117**	**0.8414**
	Delta	**0.0601**	**0.0458**	0.8793	0.9603
Without GAN	Lake	0.0581	0.0338	0.7871	0.9575
	Flood	0.1102	0.0806	0.4886	0.8084
	Delta	0.0653	0.0514	**0.8837**	**0.9655**

D. Ablation Study

We have conducted five ablation experiments here.

1) GAN architecture is ablated.
2) Both the SR module and Fusionblock are abandoned.
3) Only the SR module is added.
4) Only Fusionblock is added.
5) Both the SR module and Fusionblock are added.

The first experiment is the ablation experiment of GAN structure, and the left four are the ablation experiments of individual modules. It should be noted here that, since the training data of the farmland dataset are not enough, only the first three datasets are used here for analysis. The visual and quantitative results are shown in Figs. 21 and 22 and Tables IX and X.

1) Ablation Experiment of GAN: The first experiment is to prove the effectiveness of GAN used in our model, and the

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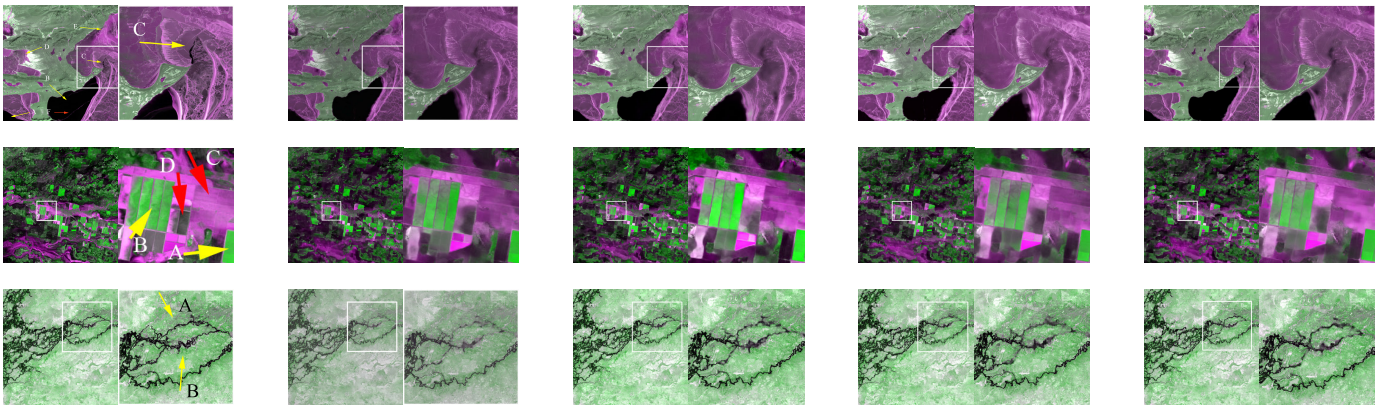


Fig. 21. Results of the ablation study. The predicted images for the three datasets are placed in three rows: the first row is for lake data, the second row is for flood data, and the third row is for delta data. The results of different ablation experiments are grouped in the same column as (a) ground truth, (b) both SR module and Fusionblock are abandoned, (c) only SR module is added, (d) only Fusionblock is added, and (e) both SR module and fusion module are added.

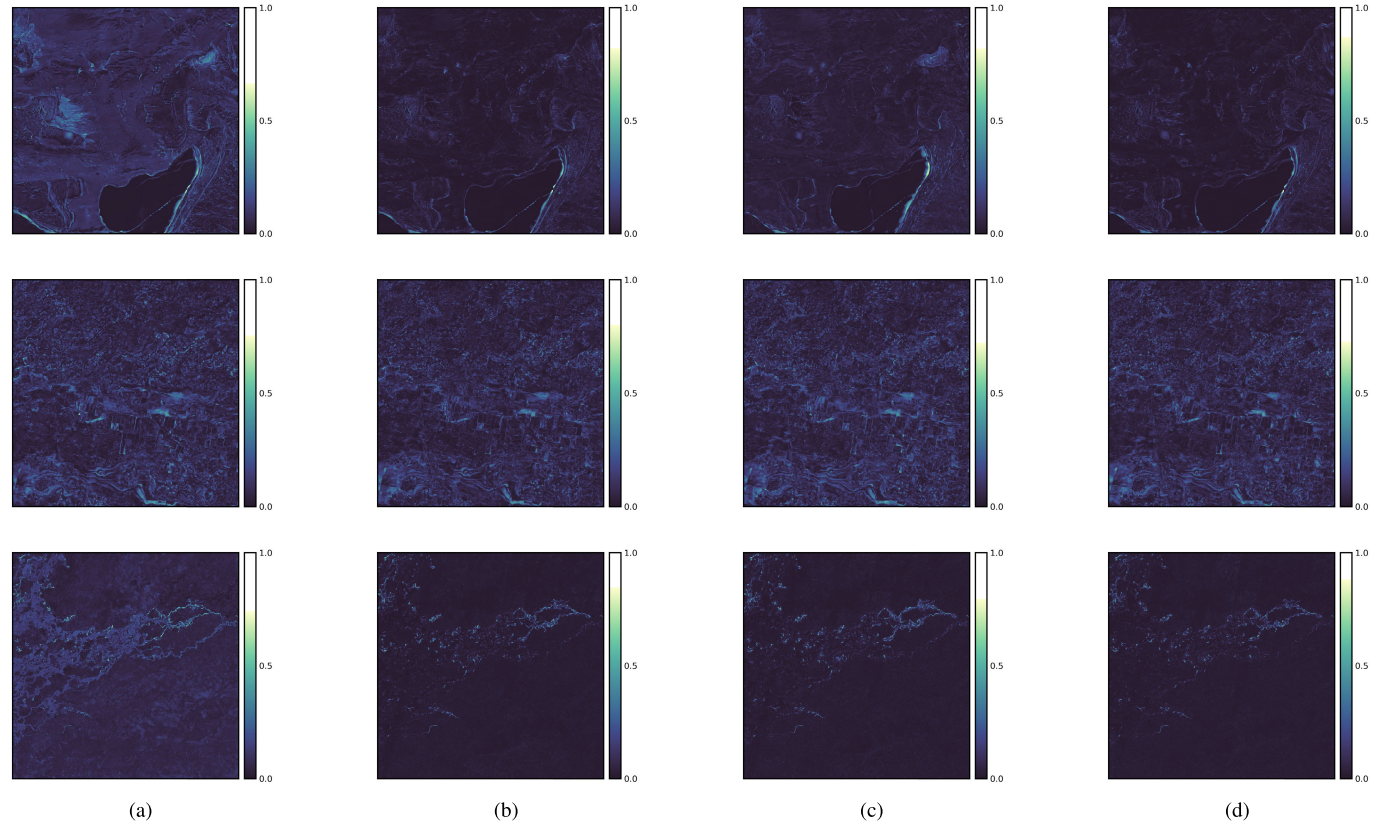


Fig. 22. Error maps of the ablation study. The error maps for three datasets are placed in three rows: the first row is for lake data, the second row is for flood data, and the third row is for delta data. The results of different ablation experiments are grouped in the same column: (a) both SR module and Fusionblock are abandoned, (b) only SR module is added, (c) only Fusionblock is added, and (d) both SR module and fusion module are added.

quantitative results are tabulated in Table IX. It can be clearly observed that, after using GAN, our model can achieve better performance on both datasets. We believe that adding the GAN loss can reduce the blurring phenomenon of the generated images and, thus, help the model recover the details of ground objects.

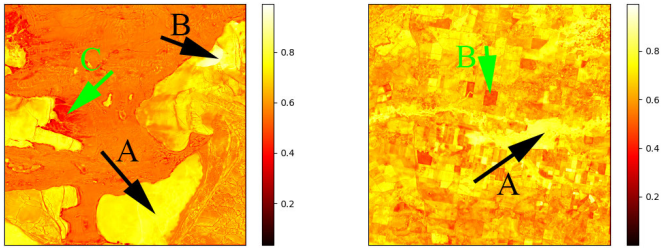
2) Ablation Experiment of Modules: Figs. 21 and 22 show the predicted images and error maps for the left four ablation experiments on modules. It can be seen that the corresponding performance on the changed areas is poor when neither SR nor Fusionblock is used. As shown in Fig. 21(b), for the lake data, the model is not able to predict the water/thin ice at Area C; for the flood data, the module is not able to predict details well on the flooding areas (Area C); and for the delta data, the model can hardly reconstruct the two narrow and meandering rivers at Areas A and B. Moreover, the tone of the predicted image deviates from the ground truth. After adding the SR module, the overall color tone of the outputs has been improved a lot and, thus, is closer to ground truths for all datasets [see Fig. 21(c)]. For example, the colors of

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TABLE X

ABLATION EXPERIMENT OF THE SR AND FUSION MODULES ON THE TWO DATASETS

Dataset	SR	Fusionblock	RSME	MAE	SSIM	UIQI
Lake	×	×	0.0937	0.0737	0.7560	0.9183
	✓	×	0.0573	0.0345	0.7869	0.9578
	×	✓	0.0691	0.0476	0.7753	0.9488
	✓	✓	**0.0555**	**0.0332**	**0.7872**	**0.9605**
Flood	×	×	0.1118	0.0819	0.4925	0.8098
	✓	×	0.1055	0.0754	0.5032	0.8217
	×	✓	0.1043	0.0761	0.5040	0.8218
	✓	✓	**0.1001**	**0.0725**	**0.5117**	**0.8414**
Delta	×	×	0.0918	0.0737	0.7953	0.8897
	✓	×	0.0503	0.0299	0.8834	0.9719
	×	✓	0.0494	0.0294	0.8808	0.9717
	✓	✓	**0.0491**	**0.0296**	**0.8853**	**0.9726**



(a) (b)

Fig. 23. Attention weight maps of (a) lake data and (b) flood data.

the land in the lake data, the flooding area (Area C) in the flood data, and the vegetation area in the delta data are more accurate. Although there may be some loss of details in the recovered image after adding the SR module, the problem of tone deviation in the predicted image can be solved.

When adding Fusionblock [see Fig. 21(d)], we can obtain different weights for regions with different degrees of variation. Fig. 23 shows the weight maps after training for the lake and flood data. It can be found that, for the lake data, the changed areas [such as Areas A and B in Fig. 23(a)] have a higher weight, while the unchanged area [such as Area C in Fig. 23(b)] has lower weights. The same situation occurs in the flood data. For Area A, where there was a large structural change due to the flood, the weight value is higher, while, for Area B, which is a farmland not affected by the flood, the weight value is lower. Therefore, the attention mechanism can provide a larger weight on $\mathbf{F}^i_{\mathrm{SSM}}$ for changed areas and a lower weight on $\mathbf{F}^i_{\mathrm{SSM}}$ for areas with small changes. That is to say, the attention mechanism can balance the weights of $\mathbf{F}^i_{\mathrm{SSM}}$ and $\mathbf{C}^i_{\mathrm{SR}}$ [as referred to (5)] as much as possible. As a result, our model can obtain more details by predicting areas with different variations more accurately. Comparing Fig. 21(c) and (d) carefully, one can find that more details can be generated when adding the Fusionblock module. For example, in the flood data, the recovered farmland is more square; and in the delta data, the generated river [Area A in Fig. 21(d)] is more continuous.

From Table X, it can be seen that our metrics can be improved whether the SR module or Fusionblock is added. Finally, when both modules are added, the model can achieve the best performance in both visual observation and quantitative metrics. The main reason is that our fusion module can better maintain the constant details in the reference image features, while the SR module can gain the changing details.

TABLE XI

ACCURACY METRICS OF THE DATA SELECT STUDY. THE UPWARD ARROW REPRESENTS AN INCREASE IN QUALITY, WHILE THE DOWNWARD ARROW REPRESENTS A DECREASE IN QUALITY

Experiment	Dataset	RMSE	MAE	SSIM	UIQI
1)	Lake	↓0.0217	↓0.0085	↓0.0082	↓0.0355
	Farmland	↓0.0158	↓0.0157	↑0.0027	↓0.087
2)	Lake	↓0.0549	↓0.0399	↓0.0681	↓0.0872
	Farmland	↓0.0323	↓0.0279	↓0.0202	↓0.1077

E. Data Selection Study

As shown in Section II-A, our data selection approach includes additional conditions to restrict the use of coarse-resolution images due to the presence of NVD and clouds in the MODIS data (see Figs. 3 and 4). Second, although MODIS captures daily images, it actually sweeps the oscillation resulting in different dates for the same area of the spot range (see Fig. 5). This can result in varying image resolutions for the same location at different dates. As illustrated in Figs. 24(b)–(d) and 25(b)–(d), due to the view angle effect, the actual spatial resolutions of the MODIS image at a specific site for different days are different. These problems will actually impact the quality of all MODIS-Landsat fusion methods. Therefore, we propose four criteria to ensure the quality of the input coarse reference images. To illustrate the importance of using high-quality coarse-resolution image pairs in the final output, we designed the following two experiments.

1) Using One High-Quality Coarse-Resolution Image: Here, we only use one high-quality coarse-resolution image as input. Without loss of generality, $\mathbf{C}_1$ is selected. The high-quality coarse-resolution image on January 17, 2018, is used for the lake data [see Fig. 24(b)], while the one on October 12, 2020, is used for the farmland data [see Fig. 25(b)]. Corresponding results are shown in Fig. 25(e) and (e). We can see that, when using only one high-quality coarse-resolution image, the performance of SRSF-GAN decreases. For the lake data, some water pixels are mistakenly predicted as ice (Area A). For the farmland data, some farmland patches covered by green vegetation are mistakenly predicted as bare land (Area A). As a result, all the accuracy metrics (including RMSE, MAE, SSIM, and UIQI) decrease for the lake and farmland data (see Table XI).

2) Using One Low-Quality Coarse-Resolution Image: Furthermore, we replace the high-quality coarse-resolution image with a low-quality one. That is, the low-quality coarse-resolution image on January 18, 2018, is used for the lake data [see Fig. 24(b)], while the one on October 13, 2020, is used for the farmland data [see Fig. 25(c)]. The results are shown in Figs. 24(f) and 25(f), respectively. We can see that SRSF-GAN has made more incorrect predictions. Meanwhile, many details are lost in the output results, which has greatly degraded

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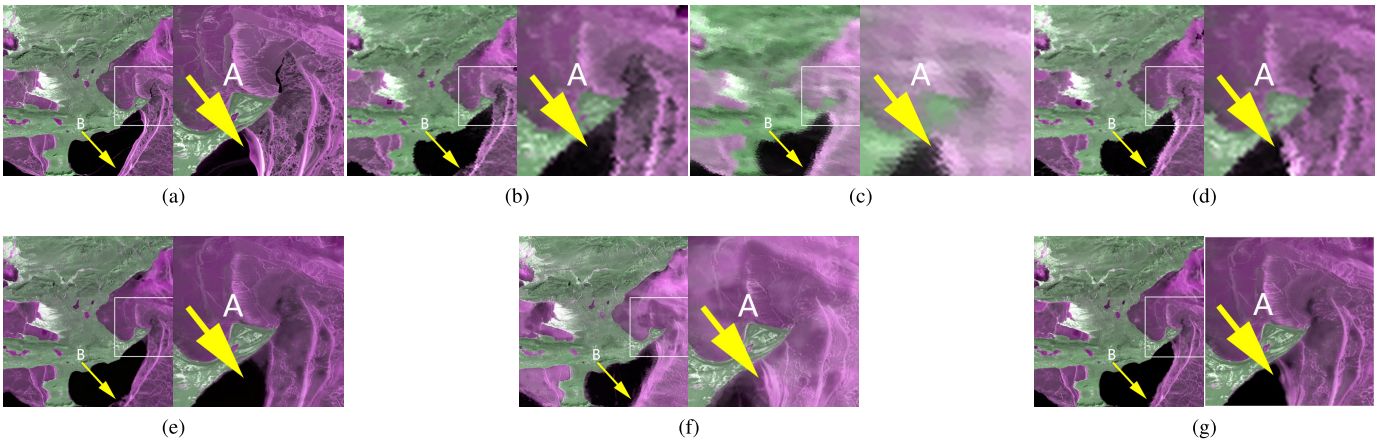


Fig. 24. Results of the data selection study on the Lake data. (a) Ground truth (Landsat image on January 18, 2018). (b) High-quality MODIS image on January 17, 2018. (c) Low-quality MODIS image on January 18, 2018. (d) High-quality MODIS image on January 19, 2018. (e) Output image of our method using one high-quality coarse-resolution image (the MODIS image on January 17, 2018). (f) Output image using one low-quality coarse-resolution image (the MODIS image on January 18, 2018). (g) Output image using two high-quality coarse-resolution images.

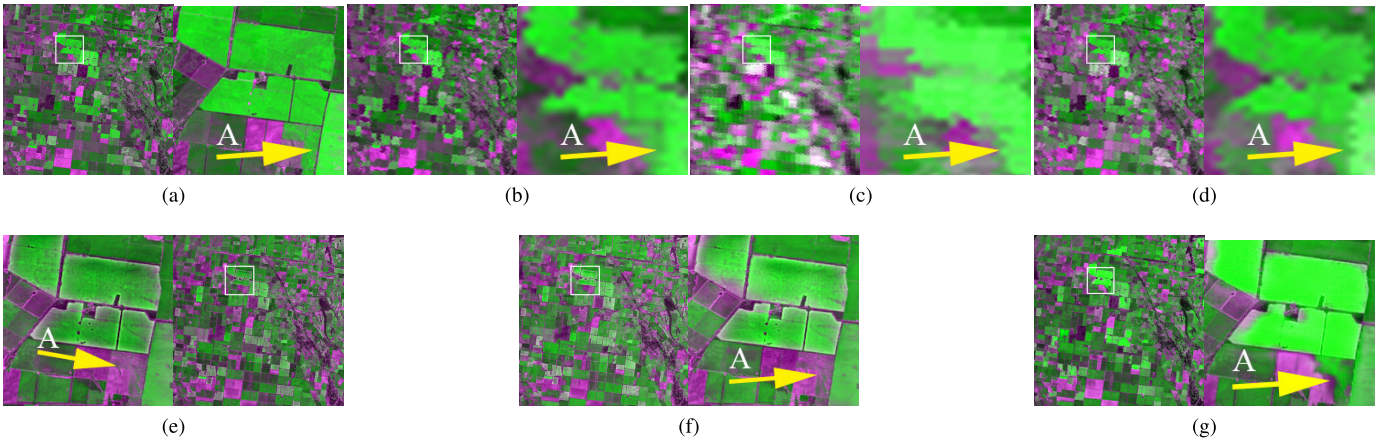


Fig. 25. Results of the data selection study on the farmland data. (a) Ground truth (Landsat image on October 13, 2020). (b) High-quality MODIS image on October 12, 2020. (c) Low-quality MODIS image on October 13, 2020. (d) High-quality MODIS image on October 14, 2020. (e) Output image of our method using one high-quality coarse-resolution image (the MODIS image on October 12, 2020). (f) Output image using one low-quality coarse-resolution image (the MODIS image on October 13, 2020). (g) Output image using two high-quality coarse-resolution images.

TABLE XII

PARAMETERS AND MACS OF FIVE DL METHODS

Model	Parameters	MACs
EDCSTFN	2.8×10^5	1.8×10^{10}
GANSTFM	(Generator) 5.8×10^5	3.8×10^{10}
	(Discriminator) 3.6×10^6	7.7×10^6
MLFF-GAN	(Generator) 5.9×10^6	1.4×10^{10}
	(Discriminator) 2.8×10^6	3.8×10^9
SWINSTFM	3.75×10^7	2.8×10^{10}
SRSF-GAN	(Generator) 3.8×10^6	1.6×10^{11}
	(Discriminator) 5.9×10^5	3.3×10^9

the quality of the generated images. Correspondingly, all the accuracy metrics further decrease for both datasets.

The two experiments indicate the importance of the data quality of the input coarse-resolution data to the final output. Therefore, it is necessary to apply data selection criteria during the training process to increase the reliability of the training. However, we must acknowledge that our data selection criteria may fail in some cases. For example, if there is a drastic land cover change within one day, two coarse images selected by our criteria may not be spatially similar to the ground truth at the prediction time, which will result in inaccurate predictions. In such cases, we recommend that users manually select appropriate coarse-resolution images for SRSF-GAN. Besides, in case we cannot obtain valid coarse and/or fine reference image pairs due to the high intensity of cloud cover, other techniques, such as performing cloud restoration [52], can be used to obtain reliable input pairs. Finally, we can also improve the quality of poor-quality pixels by applying temporal filtering [53].

F. Computational Load Comparison

The numbers of parameters and MACs are tabulated in Table XII, where MACs are the number of multiply–accumulate operations required by the network. Both metrics are used to evaluate the computation load. Due to the use of the SR module for coarse-resolution images, our model will have more MACs compared to the other four models. However, our model's performance is relatively superior to the other four models, and the number of parameters is also less than MLFF-GAN and SWINSTFM.

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IV. Conclusion

In this article, we propose a new fusion method for satellite images with different spatial and temporal resolutions, named SRSF-GAN. Unlike most spatiotemporal fusion methods, our method is spatial-based, not phase-based, which only requires coarse-resolution images at the prediction time and a fine reference image. Moreover, our method proposes a data selection step, guaranteeing the selection of high-quality coarse images and a clear fine-resolution reference that has the most spatial–spectral similarity to the coarse images at the prediction time. We have added an SR module to directly predict the fine-resolution image at the prediction time and then design a multilevel fusion module to fuse the predicted SR and fine reference images with weights automatically calculated by a spatial attention mechanism. Four comparison experiments with state-of-the-art methods show that our new framework can deal with a variety of changing types with the best performance. Besides, we have conducted a series of ablation experiments, which proved that the SR module plays a crucial role in generating details for changed areas, and the fusion module has a good integration effect for fine reference image features and SR image features. In addition, we have conducted the effectiveness of our proposed data selection criteria on coarse-resolution images. In the future, we will focus on using our proposed methods on large-scale data fusing for different applications.

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